



Testing high-dimensional regression coefficients in linear models via Bayes factor

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Abstract

This paper considers the problem of testing for the overall significance of a large number of remaining predictors in high-dimensional linear models, given that a few predictors are known to have significant effect on the response. We propose a novel test based on Bayes factor, assuming that the errors are normally distributed. The proposed test is also applicable to testing the global significance of the linear models. The asymptotic normality of the Bayes factor-based test statistic under the null hypothesis and the local alternatives is established. Furthermore, we derive the asymptotic local power function of the proposed test and compare it with that of existing methods. Monte Carlo simulation results show that the limiting approximation to the null distribution is highly accurate in finite samples, and the proposed test outperforms several well-known tests in the literature in terms of Type I error rate and the empirical power. The practical effectiveness of the proposed test is illustrated through two real-data applications.

Keywords Asymptotic normality · Bayes factor · Fixed design · Hypotheses testing · High-dimensional linear model

1 Introduction

High-dimensional data are frequently encountered in many research areas such as genomics, medical images, financial studies. Hypothesis testing in such data is a fundamentally important problem that has garnered increasing attention in recent years. Conventional testing methods is no longer applicable because the data dimension p is greater than the sample size n . In this paper, we propose a novel method based on Bayes factor for testing the regression coefficients in high-dimensional linear models with fixed design matrix.

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Consider the linear regression model of the following form

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\epsilon}, \quad (1)$$

where $\mathbf{y} \in \mathbb{R}^n$ is a response vector, $\mathbf{X}_1$ and $\mathbf{X}_2$ are $n \times q$ and $n \times p$ design matrices, respectively, $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are q - and p -dimensional vector of unknown regression coefficients, respectively. $\boldsymbol{\epsilon} \in \mathbb{R}^n$ is the vector of random errors and we assume that $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \phi^{-1}\mathbf{I}_n)$, ϕ is a positive constant in this paper. We assume that p is much greater than n and n is much greater than q . Given that a small subset of covariates, $\mathbf{X}_1$, is known to affect the response, a natural question is whether the remaining covariates also have effects on the response. It is of interest to test the following hypotheses:

$$\mathcal{H}_0 : \boldsymbol{\beta}_2 = \mathbf{0} \quad \text{versus} \quad \mathcal{H}_1 : \boldsymbol{\beta}_2 \neq \mathbf{0}. \quad (2)$$

For testing hypotheses (2), in low-dimensional settings, a classical test statistic is the F -test, which has some optimal properties (Lehmann and Romano 2005) and can be considered a likelihood ratio statistic if errors follow a normal distribution. However, even when $p < n$, Zhong and Chen (2011) showed that the F -test is adversely affected by the increasing dimension of the covariates. Relaxing the normality assumption, Wang and Cui (2013) proposed a generalized F -test under the case of $p/n \rightarrow \rho \in (0, 1)$ and established its asymptotic normality. The aforementioned two methods are not applicable in $p > n$. Recently, several tests for hypotheses of all regression coefficients have been proposed under the setting $p > n$. Zhong and Chen (2011) proposed a test statistic based on a fourth-order U -statistic, and derived its asymptotic distributions under both the null hypothesis and local alternatives. Wang and Cui (2015) extended the U -statistic-based test of Zhong and Chen (2011) to test part of regression coefficients. Moreover, Cui et al. (2018) proposed a test based on an estimated U -statistic of order two, employing the Refitted Cross-Validation (RCV) method to reduce the bias of the sample variance. Building on this work, Guo et al. (2022) introduced a two-stage conditional U -statistic test. This test incorporates a screening procedure for testing partial significance in high-dimensional linear models and employs the RCV strategy to enhance empirical power. By transforming the high-dimensional regression problem into a one-sample mean testing problem, Zhao et al. (2024) proposed a statistic composed of a U -statistic term and a power-enhancement term designed to leverage correlations among predictors. Yang et al. (2024) investigated a U -statistic-based test derived from the score function to test the significance of an high-dimensional sub-vector of regression coefficients and derived its limiting distributions. The previously mentioned methods are all U -statistic-based. However, several other classes of tests exist for the $p > n$ setting. Based on non-parametric statistics, Xu and Zhou (2021) proposed a maximum-type test using Wilcoxon scores, while Xu (2017) developed a procedure under a rank-based framework. Both of these tests are robust against outlying observations and heavy-tailed errors. Another approach, based on random projection, was proposed by Liu et al. (2023). It reduces the data dimensionality and then constructs an F -statistic in the lower-dimensional space. A further line of research builds on conditional-mean-dependence metrics to assess the effects of covariates on

the response without imposing a specific model form. In particular, Zhang et al. (2018) introduced a test for conditional mean and quantile dependence in high dimension based on the martingale difference divergence, and Li et al. (2023) proposed a test for the effects of high-dimensional covariates by aggregating cumulative covariances. More recently, Xu et al. (2025) developed center-outward ranks and signs for testing conditional quantile independence. These metric-based procedures provide a flexible and robust alternative to model-based tests, and complement the methods considered in this paper. All these methods fall under the frequentist framework, another important class of procedures for hypothesis (2) is the Bayesian approach. Goeman et al. (2006) discussed a score test on a hyperparameter in an empirical Bayesian model and shown the test had optimal power on average in a neighborhood of the null hypothesis. Goeman et al. (2011) extended the test by Goeman et al. (2006) in generalized linear model. Denote the rank decomposition of $\tilde{\mathbf{H}}_1 = \mathbf{I}_n - \mathbf{X}_1(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top$ by $\mathbf{U}_1 \mathbf{U}_1^\top$, where $\mathbf{U}_1$ is an $n \times (n - q)$ column orthogonal matrix. Then, the test statistic by Goeman et al. (2006) for hypotheses (2) is

$$T = \frac{\mathbf{y}^\top \mathbf{U}_1 \mathbf{U}_1^\top \mathbf{X}_2 \mathbf{X}_2^\top \mathbf{U}_1 \mathbf{U}_1^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{U}_1 \mathbf{U}_1^\top \mathbf{y}}. \quad (3)$$

In contrast, Wang and Xu (2021) proposed the following test statistic

$$T = -\frac{\mathbf{y}^\top \mathbf{U}_1 (\mathbf{U}_1^\top \mathbf{X}_2 \mathbf{X}_2^\top \mathbf{U}_1)^{-1} \mathbf{U}_1^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{U}_1 \mathbf{U}_1^\top \mathbf{y}}, \quad (4)$$

which represents the limit of Bayes factor, rather than the Bayes factor itself. Note that the test by Wang and Xu (2021) is designed for large signal magnitudes ($\|\beta_2\|$), while the test by Goeman et al. (2006) focused on small ones. However, focusing only on these two extremes can be limiting in practice. In this paper, we propose a different prior distribution for β_2 from those used in their work, which we introduce in Sect. 2. Besides Bayesian hypothesis testing, model selection based on the Bayes factor has also received extensive attention (see, e.g., Moreno et al. 2010; Wang and Sun 2014; Maruyama and George 2011; Johnson 2008; Maruyama and Strawderman 2014).

To the best of our knowledge, while many Bayesian tests have been proposed for hypothesis (2) in low-dimensional settings (see, e.g., Zhou and Guan 2018; Shively and Walker 2018; Goddard and Johnson 2016), most fail in high-dimensional setting. Moreover, no literature focuses on testing regression coefficients via Bayes factor in linear models under “large p , small q ” setting. This problem is challenging, as Wang and Xu (2021) proved that no nontrivial unbiased test procedure exists for fixed design matrix in this setting, although such procedures do exist for random designs. On the other hand, the likelihood function is unbounded for $p > n$, however, the Bayes factor (Kass and Raftery 1995) is still well-defined in this case. Inspired by these, in this paper, we consider testing whether a large number of predictors have significant effect on the response, given that a few predictors are known to be significant, using Bayes factor.

There are two main contributions in this paper. (i) We propose a novel, computationally efficient test based on Bayes factor. We establish its asymptotic normality under both the null hypothesis and local alternatives. Furthermore, we derive the asymptotic power function of the proposed test and compare it with that of the existing ones. Our analysis shows that the proposed test outperforms the one by Wang and Xu (2021) in the $p > n - q$ setting, as it can detect signals from a wider range of directions. Furthermore, in the $p < n - q$ setting, our test has nontrivial power and is more powerful than the test by Goeman et al. (2006). Extensive numerical experiments further confirm the superiority of our proposed test. (ii) We construct a unified framework of Bayes factor-based tests for regression coefficients in linear models, which accommodates various relationships between p and n , including the settings where $p/(n - q) \rightarrow c \in (1, \infty]$, $p/(n - q) \rightarrow c \in (0, 1)$ and $p/(n - q) \rightarrow 0$ while p grows with n . The test statistic for the third setting, which is more simplified under certain conditions, is detailed in the Supplementary material.

The remainder of the paper is organized as follows. In Sect. 2, we construct a test based on Bayes factor. In Sect. 3, we establish the asymptotic distribution of the test statistic under the null hypothesis and the local alternatives. The asymptotic power function of the proposed test is also derived and is compared with the existing ones. In Sect. 4, extensive simulation studies are conducted to compare the finite sample performances between the proposed test and some existing known ones. In Sect. 5, two real-data examples are provided to illustrate the practical utility of our test. Section 6 includes this paper. The proofs of the all theoretical results are provided in Appendix.

2 Motivations and the test statistic

The likelihood ratio test is an important test statistic in hypotheses testing. However, in high-dimensional linear models in which the dimension of parameters is greater than the sample size, the likelihood function is unbounded, which causes the likelihood ratio test to be undefined (Bai and Saranadasa 1996). Nevertheless, the Bayes factor (Kass and Raftery 1995) is still well-defined in this case. In Bayesian framework, Bayes factor is a potential tool for testing problem and model selection. As far as we know, there is no existing literature that focuses on testing regression coefficients via Bayes factor in linear models under “large p , small q ” setting. Motivated by these, we try to construct the test statistics for hypotheses (2) based on Bayes factor. The Bayes factor for hypotheses (2) is

$$\text{BF}_{10} = \frac{\int \int f_1(y|\beta_2, \beta_1, \phi) \pi_1(\beta_2, \beta_1, \phi) d\beta_1 d\beta_2 d\phi}{\int \int f_0(y|\beta_1, \phi) \pi_0(\beta_1, \phi) d\beta_1 d\phi}. \quad (5)$$

where $f_0(y|\beta_1, \phi)$ and $f_1(y|\beta_1, \beta_2, \phi)$ are the density functions for y under $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively. $\pi_0(\beta_1, \phi)$ and $\pi_1(\beta_2, \beta_1, \phi)$ are the prior distributions under $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively. If BF_{10} is large, we will reject $\mathcal{H}_0$. In Bayesian framework, the choice of prior plays an important role on the behavior of Bayes factor. The well-known g prior and its variations have been widely studied (see, e.g., Liang et al. 2008; Zhang et al. 2008; Li and Clyde 2018; Kang and Jeong 2024). Another prior is so-called

intrinsic prior (Berger and Pericchi 1996). However, these priors are not well defined in “large p , small q setting”. In the absence of prior knowledge, noninformative priors are preferred for the unknown parameters (Kass and Vaidyanathan 1992). However, these priors are often improper and may result in the Bayes factor being undefined. To avoid this pitfall, we only employ the improper priors for the “common” parameters that appears in both models under hypotheses (2) and a proper prior for the parameter that only appears in the model under $\mathcal{H}_1$. Specifically, we adopt the reference prior $\pi_0(\boldsymbol{\beta}_1, \phi) = c/\phi$, where c is a constant. Under $\mathcal{H}_1$, consider the same prior as in $\mathcal{H}_0$ for $\boldsymbol{\beta}_1$ and ϕ , that is, $\pi_1(\boldsymbol{\beta}_1, \phi) = \pi_0(\boldsymbol{\beta}_1, \phi)$. Let $\tilde{\mathbf{X}}_2 = \mathbf{U}_1^\top \mathbf{X}_2$. Denote the singular value decomposition of $\tilde{\mathbf{X}}_2$ by $\tilde{\mathbf{U}}_2 \tilde{\mathbf{\Lambda}}_2 \tilde{\mathbf{V}}_2^\top$, where $\tilde{\mathbf{U}}_2, \tilde{\mathbf{V}}_2$ are $(n-q) \times (n-q)$ and $p \times (n-q)$ column orthogonal matrices, respectively, and $\tilde{\mathbf{\Lambda}}_2 = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_{n-q}})$ with $\lambda_i, i = 1, \dots, n-q$ is the i th largest eigenvalue of $\tilde{\mathbf{X}}_2 \tilde{\mathbf{X}}_2^\top$. Let $\tilde{\mathbf{P}} = (\tilde{\mathbf{V}}_2^\top \mathbf{P})$, where $\mathbf{P}$ is $p \times (p - (n-q))$ column orthogonal matrix. We consider the following simple prior $\pi_1(\boldsymbol{\beta}_2 | \boldsymbol{\beta}_1, \phi) \sim N(\mathbf{0}, \phi^{-1} \tilde{\mathbf{P}} \boldsymbol{\Gamma}^{-1} \tilde{\mathbf{P}}^\top)$, where $\boldsymbol{\Gamma} = \text{diag}(\gamma_1, \gamma_2, \dots, \gamma_p)$ with $\gamma_i, i = 1, 2, \dots, p$ satisfying $0 < \gamma_i < \infty$.

Remark 1 Since $\mathbf{X}_2^\top \mathbf{X}_2$ is not invertible when $p > n$, the g prior is not applicable. The strategy of Wang and Xu (2021) is to simply adopt a positive hyperparameter $\kappa, \kappa > 0$. They used the prior as $\pi_1(\boldsymbol{\beta}_2 | \boldsymbol{\beta}_1, \phi) \sim N(\mathbf{0}, \phi^{-1} \kappa^{-1} \mathbf{I}_p)$. Under this prior distribution, the part that works in (8) is

$$T_\kappa = \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Lambda}}_2 (\tilde{\mathbf{\Lambda}}_2^2 + \kappa \mathbf{I}_{n-q})^{-1} \tilde{\mathbf{\Lambda}}_2 \tilde{\mathbf{U}}_2^\top \tilde{\mathbf{y}}}{\tilde{\mathbf{y}}^\top \tilde{\mathbf{y}}}. \quad (6)$$

The hyperparameter κ is unknown and requires determination. Wang and Xu (2021) think the global power behavior is more meaning and thus let $\kappa \rightarrow 0$. The test proposed by Wang and Xu (2021) is exactly $\lim_{\kappa \rightarrow 0} \frac{T_\kappa - 1}{\kappa}$, as shown in (4). Contrary to their strategy, Goeman et al. (2006) consider small $\|\boldsymbol{\beta}_2\|$, the limit $\lim_{\kappa \rightarrow \infty} \kappa T_\kappa$ is exactly the test statistics of Goeman et al. (2006), as shown in (3). Both of their tests are frequentist test statistics not are the Bayes factor-based tests and they do not give specific scheme to select κ instead of adopting the limit. In practice, the choice of an appropriate κ is not feasible. Nevertheless, in this paper, the proposed prior distribution for $\boldsymbol{\beta}_2$ has a simple form, yet it is both flexible and informative for at least for two reasons.

(i) The asymptotic properties of Bayes factors are challenging to derive because of the presence of ratio of quadratic forms. However, adopting this simple prior form makes the order of the key matrix $\tilde{\boldsymbol{\Gamma}}$ explicit. This, in turn, renders it tractable to determine the orders of the expectations and variances of the pivotal ratio of quadratic forms T (defined in (9)), thereby facilitating the derivation of the asymptotic distribution of our test statistic, T_{BF} defined in (13).

(ii) It is not only applicable under $p > n - q$ but also $p \leq n - q$ settings. Furthermore, if we set $\gamma_i = \kappa$ for all $i = 1, \dots, p$, the prior distribution placed on $\boldsymbol{\beta}_2$ reduces to the one proposed by Wang and Xu (2021), which is therefore a special case of our proposed prior.

In dense signal scenarios, where widespread weak signals are often present, an adaptive shrinkage prior is required to accommodate these signals. To this end, and

motivated by Li and Clyde (2018) and Maruyama and George (2011), we recommend setting $\gamma_i = p$ in our simulations. Notice that our theoretical results apply to the general case, as long as γ_i satisfies $0 < \gamma_i < \infty, i = 1, \dots, p$.

Accordingly, we obtain the prior distribution, $\pi_1(\boldsymbol{\beta}_2, \boldsymbol{\beta}_1, \phi) = \pi_1(\boldsymbol{\beta}_2 | \boldsymbol{\beta}_1, \phi) \pi_1(\boldsymbol{\beta}_1, \phi)$. From (5), the Bayes factor in favor of $\mathcal{H}_1$ against $\mathcal{H}_0$, after taking its logarithm and multiplying by 2, is given analytically by

$$2 \log \text{BF}_{10} = \log |\tilde{\mathbf{P}} \boldsymbol{\Gamma} \tilde{\mathbf{P}}^\top| - \log |\mathbf{X}_2^\top \tilde{\mathbf{H}}_1 \mathbf{X}_2 + \tilde{\mathbf{P}} \boldsymbol{\Gamma} \tilde{\mathbf{P}}^\top| \\ - (n - q) \log \left\{ 1 - \frac{\mathbf{y}^\top \tilde{\mathbf{H}}_1 \mathbf{X}_2 (\mathbf{X}_2^\top \tilde{\mathbf{H}}_1 \mathbf{X}_2 + \tilde{\mathbf{P}} \boldsymbol{\Gamma} \tilde{\mathbf{P}}^\top)^{-1} \mathbf{X}_2^\top \tilde{\mathbf{H}}_1 \mathbf{y}}{\mathbf{y}^\top \tilde{\mathbf{H}}_1 \mathbf{y}} \right\}. \quad (7)$$

Let $\tilde{\mathbf{y}} = \mathbf{U}_1^\top \mathbf{y}$. Then (7) can be rewritten as

$$2 \log \text{BF}_{10} = \sum_{i=1}^{n-q} \log \gamma_i - \sum_{i=1}^{n-q} \log (\gamma_i + \lambda_i) - (n - q) \log (1 - T), \quad (8)$$

where

$$T = \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{U}}_2 \tilde{\mathbf{F}} \tilde{\mathbf{U}}_2^\top \tilde{\mathbf{y}}}{\tilde{\mathbf{y}}^\top \tilde{\mathbf{y}}} \quad \text{with} \quad \tilde{\mathbf{F}} = \text{diag} \left(\frac{\lambda_1}{\gamma_1 + \lambda_1}, \dots, \frac{\lambda_{n-q}}{\gamma_{n-q} + \lambda_{n-q}} \right). \quad (9)$$

Let the subscript “0” denote that the operation on T is taken under the null hypotheses. According to Bao and Kan (2013), through some calculations,

$$\begin{aligned} \mathbb{E}_0(T) &= \text{tr}(\tilde{\mathbf{F}})/(n - q), \\ \mathbb{E}_0(T^2) &= \frac{2\text{tr}(\tilde{\mathbf{F}}^2) + (\text{tr}(\tilde{\mathbf{F}}))^2}{(n - q)(n - q + 2)}, \\ \text{Var}_0(T) &= \frac{2 \left(\text{tr}(\tilde{\mathbf{F}}^2) - (\text{tr}(\tilde{\mathbf{F}}))^2/(n - q) \right)}{(n - q)(n - q + 2)}. \end{aligned} \quad (10)$$

For convenience, denote $\mathbb{E}_0(T)$, $\mathbb{E}_0(T^2)$ and $(n - q)(n - q + 2)\text{Var}_0(T)$ by μ_0 , μ_{02} and σ_0^2 , respectively. By Taylor expansion, we have

$$\begin{aligned} 2 \log \text{BF}_{10} &- \sum_{i=1}^{n-q} \log \gamma_i + \sum_{i=1}^{n-q} \log (\gamma_i + \lambda_i) \\ &= (n - q) \left(-\log(1 - \mu_0) + \frac{T - \mu_0}{1 - \mu_0} + \frac{(T - \mu_0)^2}{2(1 - \mu_0)^2} \right) + o_p(1) \\ &= T_1 + T_2 + o_p(1) \end{aligned} \quad (11)$$

where

$$\begin{aligned} T_1 &= (n - q) \left(\frac{(T - \mu_0)(1 - 2\mu_0)}{(1 - \mu_0)^2} + \frac{T^2 - \mu_0^2}{2(1 - \mu_0)^2} \right), \\ T_2 &= (n - q) \left(-\log(1 - \mu_0) + \frac{\text{Var}_0(T)}{2(1 - \mu_0)^2} \right). \end{aligned} \quad (12)$$

Note that the term T_2 is nonrandom. We propose our test statistic as

$$T_{\text{BF}} = \frac{2 \log \text{BF}_{10} - \sum_{i=1}^{n-q} \log \gamma_i + \sum_{i=1}^{n-q} \log(\gamma_i + \lambda_i) + (n - q) \left(\log(1 - \mu_0) - \frac{\text{Var}_0(T)}{2(1 - \mu_0)^2} \right)}{\sqrt{\sigma_0^2 / (1 - \mu_0)}} \quad (13)$$

Remark 2 There is no explicit relationship imposed between the growth rates of p and n , which makes the tests applicable across a wide range of high-dimensional settings, for example, as $n \rightarrow \infty$, $n - q \rightarrow \infty$, $p/(n - q) \rightarrow c$, where c is a constant greater than 1, or $p/(n - q) \rightarrow \infty$. We denote p and q as p_n and q_n , respectively, to emphasize their dependence on n . We assume that $p_n \gg n$, $n \gg q_n$.

Remark 3 Because $1 - \mu_0 \in (0, 1)$, the test T_{BF} is well defined. Furthermore, the test statistic (13) remains valid when $p_n/(n - q_n) \rightarrow c \in (0, 1)$, even though our primary focus is on the case where $c \in (1, \infty)$. The main adjustments for the $c \in (0, 1)$ case lie in the dimensions of several components. Specifically, the upper limit of the summations, such as in $\sum_i \log \gamma_i$ and $\sum_i \log(\lambda_i + \gamma_i)$, should be p_n , not $n - q_n$. In addition, $\tilde{\Gamma}$ is a $p_n \times p_n$ matrix.

To formulate a test procedure based on (13), its asymptotic distribution must be obtained. We will present the main results in the next section.

3 Main results

In this section, we investigate the asymptotic normality of the test statistic T_{BF} under the null hypothesis and the local alternatives and derive the asymptotic local power function of T_{BF} . In spite of these asymptotic properties are derived under the $p_n/(n - q_n) \rightarrow c \in (1, \infty]$ setting, they are still applicable for the $p_n/(n - q_n) \rightarrow c \in (0, 1)$ setting. It should be noted that when $p_n/(n - q_n) \rightarrow 0$, under some certain conditions, the test statistic based on the Bayes factor differs from its high-dimensional counterpart, as these conditions conflict with the assumption of $p_n/(n - q_n) \rightarrow c \in (1, \infty)$. We aim to construct unified Bayes factor-based tests for regression coefficients in linear models that can accommodate many relationships between p_n and n . Owing to space limitations, the test procedures for the case where $p_n/(n - q_n) \rightarrow 0$ while p_n diverges are presented in the Supplementary material.

To investigate the asymptotic distribution of T_{BF} , we give the following lemma and proposition.

Lemma 1 *Under the following condition*

$$\frac{\max_{1 \leq i \leq n-q_n} \left((n-q_n) \frac{\lambda_i}{\gamma_i + \lambda_i} - \text{tr}(\tilde{\Gamma}) \right)^2}{(n-q_n)^2 \sigma_0^2} \rightarrow 0, \quad (14)$$

as $n \rightarrow \infty$, $n - q_n \rightarrow \infty$, for any $x \in \mathbb{R}$, we have

$$\begin{aligned} & \Pr \left((n-q_n) \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{U}}_2 \tilde{\Gamma} \tilde{\mathbf{U}}_2^\top \tilde{\mathbf{y}}}{\tilde{\mathbf{y}}^\top \tilde{\mathbf{y}}} \leq \zeta_x \right) \\ &= \Phi \left(\frac{\left(\left(1 + \frac{\phi \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \tilde{\mathbf{V}}_2^\top \beta_2}{n-q_n} \right) x - \phi \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \left(\frac{\tilde{\mathbf{r}} - \frac{\text{tr}(\tilde{\Gamma})}{n-q_n} \mathbf{I}_{n-q_n}}{\sqrt{\sigma_0^2}} \right) \tilde{\mathbf{V}}_2^\top \beta_2 \right)}{\sqrt{1 + 4\phi \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \left(\frac{\tilde{\mathbf{r}} - \frac{\zeta_x}{n-q_n} \mathbf{I}_{n-q_n}}{\sqrt{\sigma_0^2}} \right)^2 \tilde{\mathbf{V}}_2^\top \beta_2}} \right) + o(1), \end{aligned} \quad (15)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution, $\mathbf{I}_{n-q_n}$ is the $(n-q_n) \times (n-q_n)$ identity matrix and

$$\zeta_x = \text{tr}(\tilde{\Gamma}) + \sqrt{\sigma_0^2} x.$$

Remark 4 Condition (14) is used to prove that $\lambda_1(\mathbf{B}^2)/\text{tr}(\mathbf{B}^2) \rightarrow 0$, where $\mathbf{B} = (n-q_n)\tilde{\mathbf{U}}_2\tilde{\Gamma}\tilde{\mathbf{U}}_2^\top - \zeta_x\mathbf{I}_{n-q_n}$ is defined as in the proof of Lemma 1 where $\lambda_1(A)$ denotes the largest eigenvalue of a matrix A . $\lambda_1(\mathbf{B}^2)/\text{tr}(\mathbf{B}^2) \rightarrow 0$ is a widely used condition for deriving the asymptotic normality of quadratic forms (see, e.g., Bai and Saranadasa 1996; Zhong and Chen 2011). Condition (14) clearly holds. Specifically, the diagonal elements of the matrix $\tilde{\Gamma}$, given by $\lambda_i/(\gamma_i + \lambda_i)$, are all in the interval $(0, 1)$, which implies that $\text{tr}(\tilde{\Gamma}) = O(n-q_n)$. Similarly, $\sigma_0^2 = O(n-q_n)$. Therefore the numerator in (14) is of order $O((n-q_n)^2)$ while the denominator is of order $O((n-q_n)^3)$. This is a straightforward result from the prior placed on β_2 . Furthermore, the prior for β_2 makes it tractable to determine the orders of the expectations and variances of certain quantities related to the key quadratic form ratio, T . This tractability, in turn, facilitates the derivation of the asymptotic distribution of our test statistic, T_{BF} .

Remark 5 Since no conditions are imposed on β_2 , Lemma 1 establishes the asymptotic distribution of $(n-q_n)T$ under not only the null hypothesis but also global alternatives.

To facilitate the analysis of the asymptotic local power of the proposed test, we write $\tilde{\mathbf{V}}_2^\top \beta_2 = (\omega_1, \omega_2, \dots, \omega_{n-q_n})^\top$, where ω_i denotes the coefficient corresponding to the projection of β_2 onto the i -th eigenvector of $\tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2$, for $i = 1, 2, \dots, n-q_n$. As stated in Lemma 1, the asymptotic behavior of T under the global alternative is complicated, which makes it difficult to analyze the asymptotic power functions of

T_{BF} . Therefore, we next derive a simplified expression for the asymptotic distribution of T by imposing local alternative conditions.

Proposition 1 *Suppose that condition (14) holds and that the local alternative has the following form*

$$\beta_2^\top \tilde{V}_2 \tilde{\Lambda}_2^2 \tilde{V}_2^\top \beta_2 = O\left((n - q_n)^{1/2}\right) \text{ and } \max_{1 \leq i \leq n - q_n} (\lambda_i \omega_i^2) = o(1). \quad (16)$$

Then, as $n \rightarrow \infty$, $n - q_n \rightarrow \infty$, for any $x \in \mathbb{R}$

$$\Pr\left((n - q_n) \frac{\tilde{y}^\top \tilde{U}_2 \tilde{\Gamma} \tilde{U}_2^\top \tilde{y}}{\tilde{y}^\top \tilde{y}} \leq \zeta_x\right) = \Phi\left(x - \frac{\phi \beta_2^\top \tilde{V}_2 \tilde{\Lambda}_2^2 \left(\tilde{\Gamma} - \frac{\text{tr}(\tilde{\Gamma})}{n - q_n} \mathbf{I}_{n - q_n}\right) \tilde{V}_2^\top \beta_2}{\sqrt{\sigma_0^2}}\right) + o(1). \quad (17)$$

Remark 6 Note that the local alternative $\beta_2^\top \tilde{V}_2 \tilde{\Lambda}_2^2 \tilde{V}_2^\top \beta_2 = O\left((n - q_n)^{1/2}\right)$ implies that $\beta_2^\top \tilde{V}_2 \tilde{\Lambda}_2^2 \tilde{\Gamma}^i \tilde{V}_2^\top \beta_2 = O\left((n - q_n)^{1/2}\right)$, for $i = 1, 2$. This implication holds because all elements of the matrix $\tilde{\Gamma}$ are greater than 0 and less than 1.

By virtue of Lemma 1 and Proposition 1, the test statistic T_{BF} has the following property.

Theorem 1 *Suppose that condition (14) holds. Then as $n \rightarrow \infty$, $n - q_n \rightarrow \infty$, the test statistic defined in (13) has the following properties*

(i) *Under the null hypothesis*

$$\frac{2 \log \text{BF}_{10} - \sum_{i=1}^{n - q_n} \log \gamma_i + \sum_{i=1}^{n - q_n} \log (\gamma_i + \lambda_i) + (n - q_n) \left(\log(1 - \mu_0) - \frac{\text{Var}_0(T)}{2(1 - \mu_0)^2} \right)}{\sqrt{\sigma_0^2 / (1 - \mu_0)}} \xrightarrow{d} N(0, 1) \quad (18)$$

where “ $\xrightarrow{d}$ ” stands for convergence in distribution.

(ii) *Under the local alternatives (16)*

$$\begin{aligned} & \frac{2 \log \text{BF}_{10} - \sum_{i=1}^{n - q_n} \log \gamma_i + \sum_{i=1}^{n - q_n} \log (\gamma_i + \lambda_i) + (n - q_n) \left(\log(1 - \mu_0) - \frac{\text{Var}_0(T)}{2(1 - \mu_0)^2} \right)}{\sqrt{\sigma_0^2 / (1 - \mu_0)}} \\ & - \frac{\phi \beta_2^\top \tilde{V}_2 \tilde{\Lambda}_2^2 \left(\tilde{\Gamma} - \frac{\text{tr}(\tilde{\Gamma})}{n - q_n} \mathbf{I}_{n - q_n} \right) \tilde{V}_2^\top \beta_2}{\sqrt{\sigma_0^2}} \xrightarrow{d} N(0, 1) \end{aligned} \quad (19)$$

Theorem 1 establishes the asymptotic distribution of the test statistic T_{BF} under the null hypothesis and the local alternatives (16). From Theorem 1, we formulate a test procedure; that is, we reject $\mathcal{H}_0$ at the asymptotic significance level α if

$$2 \log \text{BF}_{10} > z_{1-\alpha} \frac{\sqrt{\sigma_0^2}}{1 - \mu_0} - (n - q_n) \left(\log(1 - \mu_0) - \frac{\text{Var}_0(T)}{2(1 - \mu_0)^2} \right) + \sum_{i=1}^{n-q_n} \log \gamma_i - \sum_{i=1}^{n-q_n} \log(\gamma_i + \lambda_i), \quad (20)$$

where $z_{1-\alpha}$ is the $1 - \alpha$ quantile of the standard normal distribution. Furthermore, we give the following asymptotic power function of the proposed test under the local alternatives in (16).

Corollary 1 *Under the local alternatives in (16), the asymptotic power function of the Bayes factor-based test is*

$$\beta_{BF} = \Phi \left(-z_{1-\alpha} + \frac{\phi \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \left(\tilde{\Gamma} - \frac{\text{tr}(\tilde{\Gamma})}{n-q_n} \mathbf{I}_{n-q_n} \right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{\sigma_0^2}} \right). \quad (21)$$

Since the ZLLZ, ZC, CGZ and GZDC tests are designed for the random design matrix, and the GVV test has been shown to detect signals from a more limited range of directions than the WX test, we focus our comparison of asymptotic local power on the proposed test and the WX test. The asymptotic local power function of the test proposed by Wang and Xu (2021) is as follows

$$\Phi \left(-z_{1-\alpha} + \frac{\phi \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \left(\tilde{\Lambda}_2^{-2} - \frac{\text{tr}(\tilde{\Lambda}_2^{-2})}{n-q_n} \mathbf{I}_{n-q_n} \right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{2 \left(\text{tr}(\tilde{\Lambda}_2^{-4}) - \frac{(\text{tr}(\tilde{\Lambda}_2^{-2}))^2}{n-q_n} \right)}} \right). \quad (22)$$

From the expressions of asymptotic local power functions of T_{BF} and the test proposed by Wang and Xu (2021), the distinction between the two tests stems from the behavior of the quantities $\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \left(\tilde{\Gamma} - \frac{\text{tr}(\tilde{\Gamma})}{n-q_n} \mathbf{I}_{n-q_n} \right) \tilde{\mathbf{V}}_2^\top \beta_2$ and $\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \left(\tilde{\Lambda}_2^{-2} - \frac{\text{tr}(\tilde{\Lambda}_2^{-2})}{n-q_n} \mathbf{I}_{n-q_n} \right) \tilde{\mathbf{V}}_2^\top \beta_2$. The signs of these quantities govern whether the tests exhibit nontrivial power. Notably, neither quantity is positive definite, a result that is consistent with the fact that no test can guarantee nontrivial power in “large p , small q ” setting (Wang and Xu 2021). Hence, it is inappropriate to compare the asymptotic power function between our test statistic and that of Wang and Xu (2021) by the asymptotic relative efficiency (ARE) (Liu et al. 2023; Zhao et al. 2024). However, we can consider the scenarios in which the quantities are positive definite. The condition $\max_{1 \leq i \leq n-q_n} \lambda_i^2 / \sum_{i=1}^{n-q_n} \lambda_i^2 \rightarrow 0$ implies the absence of dominant or “spiked” eigenvalues. This is a reasonable condition. We analyze the power behavior by taking a

geometric decay eigenvalues $\lambda_i \propto \rho^{i-1}$, $0 < \rho < 1$ as a simple example for illustrative purposes. Let $a_i = p_n/\lambda_i$ for $i = 1, \dots, n - q_n$. The set of values $\{a_i\}$ forms a right-skewed distribution. Consequently, fewer than 50% of the a_i values will be larger than their sample mean, $\sum_{j=1}^{n-q_n} a_j / (n - q_n)$. In contrast, let $b_i = 1/(1 + p_n/\lambda_i)$, the set of values $\{b_i\}$ results in a left-skewed distribution. Thus, more than 50% of the b_i values will be larger than their sample mean, $\sum_{j=1}^{n-q_n} b_j / (n - q_n)$. The analysis suggests that our proposed test, T_{BF} , is more sensitive to signals distributed across many directions than that of Wang and Xu (2021).

From the above analysis, we know that if ϵ were indeed normally distributed, then the critical value can be directly obtained. Next, we consider the errors are non-Gaussian and generalize the asymptotic normality established in Theorem 1 to certain non-Gaussian error distributions under specific moment conditions. We give the moment conditions as follows.

Condition 1. ϵ_i are iid random variable with $E(\epsilon_1) = 0$, $\text{Var}(\epsilon_1) = 1$. Furthermore, suppose the distribution of ϵ_1 is symmetric about 0 and has finite eighth moments.

Proposition 2 Denote $\text{tr}(\tilde{\Gamma})/(n - q_n)$, $2 \left(\text{tr}(\tilde{\Gamma}^2) - (\text{tr}(\tilde{\Gamma}))^2/(n - q_n) \right)$ by μ_0 and σ_0^2 , respectively and let

$$R = \frac{(n - q_n) \left[(T - \mu_0)^2 - \sigma_0^2/(n - q_n)(n - q_n + 2) \right]}{2\mu_0\sqrt{\sigma_0^2}}.$$

Then, under the null hypothesis, $E(R) = O((n - q_n)^{-1/2})$ and $\text{Var}(R) = O((n - q_n)^{-1})$.

Remark 7 Combining Proposition 2 and the proofs of Theorem 1, we obtain the following results under the null hypothesis.

$$T_{BF} = \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} + o_p(1). \quad (23)$$

We now apply Theorem 2 of Wang and Xu (2021) to approximate the null distribution of the proposed statistic T_{BF} . For an $n \times n$ matrix $\mathbf{A}_0$ and a real number τ , let $F(x; \mathbf{A}_0, \tau)$ be the cumulative distribution function of $\tau \sum_{i=1}^n a_{i,i} z_i + 2 \sum_{1 \leq i < j \leq n} a_{i,j} \tilde{z}_i \tilde{z}_j$, where $z_1, \dots, z_n, \tilde{z}_1, \dots, \tilde{z}_n$ are i.i.d. $N(0, 1)$ random variables.

Under the null hypothesis, using Theorem 2 of Wang and Xu (2021), we obtain

$$\begin{aligned} & \Pr \left(\frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} > x \right) \\ &= \Pr \left((\sqrt{\phi} \epsilon)^\top \left\{ \mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\Gamma} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top - \left(\frac{\text{tr}(\tilde{\Gamma}) + \sqrt{\sigma_0^2} x}{n - q_n} \right) \mathbf{U}_1 \mathbf{U}_1^\top \right\} (\sqrt{\phi} \epsilon) > 0 \right) \end{aligned}$$

$$\approx 1 - F \left(\sqrt{\sigma_0^2 x}; \mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top - \left(\frac{\text{tr}(\tilde{\mathbf{\Gamma}}) + \sqrt{\sigma_0^2 x}}{n - q_n} \right) \mathbf{U}_1 \mathbf{U}_1^\top, \sqrt{\phi^2 \mathbb{E}(\epsilon_1^4) - 1} \right). \quad (24)$$

Thus, in order to obtain a test with significance level α asymptotically, it is necessary to use a consistent estimator $\hat{\tau}$ of $\sqrt{\phi^2 \mathbb{E}(\epsilon_1^4) - 1}$ and solve the critical value x from the equation

$$F \left(\sqrt{\sigma_0^2 x}; \mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top - \left(\frac{\text{tr}(\tilde{\mathbf{\Gamma}}) + \sqrt{\sigma_0^2 x}}{n - q_n} \right) \mathbf{U}_1 \mathbf{U}_1^\top, \hat{\tau} \right) = 1 - \alpha.$$

For ease of implementation, we propose a one-step iterative algorithm and set the initial point to $x^{(0)} = 0$. This starting point is chosen to satisfy

$$\text{tr} \left(\mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top - \left(\frac{\text{tr}(\tilde{\mathbf{\Gamma}}) + \sqrt{\sigma_0^2 x}}{n - q_n} \right) \mathbf{U}_1 \mathbf{U}_1^\top \right) = 0.$$

Then the critical value obtained from the one step iteration is

$$x^{(1)} = F^{-1} \left(1 - \alpha; \mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top - \frac{\text{tr}(\tilde{\mathbf{\Gamma}})}{n - q_n} \mathbf{U}_1 \mathbf{U}_1^\top, \hat{\tau} \right).$$

We adopt a consistent estimator of $\sqrt{\phi^2 \mathbb{E}(\epsilon_1^4) - 1}$, which is identical to the one used in Wang and Xu (2021). The estimator is given by

$$\hat{\tau}^2 = \frac{\frac{(n - q_n)^2 \sum_{i=1}^n \tilde{\epsilon}_i^4}{(\tilde{\epsilon}^\top \tilde{\mathbf{H}}_1 \tilde{\epsilon})^2} - 3, \text{tr}(\tilde{\mathbf{H}}_1^{\circ 2})}{\left(\text{tr}(\tilde{\mathbf{H}}_1^{\circ 2}) \right)^2} + 2,$$

where $\tilde{\epsilon} = (\tilde{\epsilon}_1, \tilde{\epsilon}_2, \dots, \tilde{\epsilon}_n)^\top = \tilde{\mathbf{H}}_1 \mathbf{y}$ and $\tilde{\mathbf{H}}_1^{\circ 2} = \tilde{\mathbf{H}}_1 \circ \tilde{\mathbf{H}}_1$, where $\circ$ denotes the Hadamard product.

According to (23), we reject the null hypothesis if

$$T_{\text{BF}} > x^{(1)}.$$

This test procedure is asymptotically exact at significance level α under Condition 1.

Table 1 Empirical sizes and powers of T_{BF} , WX , GVV , $ZLLZ$, ZC , CGZ and $GZDC$ tests at 5% significance level based on 5000 independent replications for case I with $\epsilon_i \sim N(0, 1)$ and $\epsilon_i \sim (3/4)^{1/2}t_8$ under dense and sparse alternatives

Dense															
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$						$\epsilon_i \sim (3/4)^{1/2}t_8$							
		New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50,1200)	0	0.0530	0.0390	0.0556	0.0910	0.0582	0.0666	0.0640	0.0678	0.0532	0.0704	0.0832	0.0556	0.0638	0.0500
	10	0.5198	0.4624	0.4986	0.1648	0.2256	0.2538	0.1660	0.5192	0.4562	0.4964	0.1698	0.2204	0.2548	0.1620
	20	0.6056	0.5648	0.5802	0.1792	0.2500	0.2906	0.1820	0.6100	0.5536	0.5748	0.1800	0.2464	0.2860	0.1700
	30	0.6440	0.6052	0.6084	0.1832	0.2640	0.3050	0.1920	0.6444	0.6010	0.6058	0.1882	0.2614	0.3000	0.1940
(100,1200)	0	0.0476	0.0322	0.0522	0.0638	0.0442	0.0466	0.0420	0.0706	0.0606	0.0788	0.0646	0.0470	0.0494	0.0380
	10	0.7852	0.7200	0.7484	0.2026	0.3916	0.4216	0.1140	0.7812	0.7118	0.7474	0.1974	0.3858	0.4132	0.1420
	20	0.9054	0.8814	0.8606	0.2414	0.4876	0.5228	0.1380	0.9004	0.8698	0.8676	0.2350	0.4892	0.5232	0.1780
	30	0.9394	0.9272	0.8968	0.2610	0.5278	0.5646	0.1640	0.9328	0.9194	0.8998	0.2570	0.5342	0.5706	0.1780
Sparse															
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$						$\epsilon_i \sim (3/4)^{1/2}t_8$							
		New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50,1200)	10	0.2564	0.2000	0.2540	0.1290	0.1328	0.1558	0.1140	0.2714	0.2208	0.2740	0.1278	0.1328	0.1602	0.1080
	20	0.3912	0.3274	0.3842	0.1442	0.1752	0.2020	0.1420	0.3950	0.3344	0.3892	0.1416	0.1776	0.2050	0.1440
	30	0.4686	0.4090	0.4510	0.1534	0.1958	0.2258	0.1480	0.4726	0.4100	0.4526	0.1518	0.2016	0.2342	0.1820
(100,1200)	10	0.3188	0.2294	0.3250	0.1204	0.1622	0.1778	0.0660	0.3450	0.2634	0.3510	0.1224	0.1620	0.1758	0.0480
	20	0.5580	0.4530	0.5380	0.1554	0.2582	0.2840	0.0900	0.5704	0.4780	0.5538	0.1584	0.2582	0.2760	0.0800
	30	0.6944	0.6050	0.6612	0.1764	0.3364	0.3596	0.1220	0.7044	0.6184	0.6654	0.1856	0.3268	0.3538	0.0940

Table 2 Empirical sizes and powers of T_{BF} , WX, GVV, ZLLZ, ZC, CGZ and GZDC tests at 5% significance level based on 5000 independent replications for case II with $\epsilon_i \sim N(0, 1)$ and $\epsilon_i \sim (3/4)^{1/2}t_8$ under dense and sparse alternatives

Dense															
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$						$\epsilon_i \sim (3/4)^{1/2}t_8$							
		New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50, 1200)	0	0.0626	0.0550	0.0612	0.0810	0.0570	0.0658	0.0600	0.0618	0.0558	0.0628	0.0756	0.0594	0.0672	0.0740
	10	0.4878	0.4630	0.4774	0.2128	0.4518	0.4614	0.1700	0.4944	0.4640	0.4756	0.2206	0.4586	0.4652	0.2020
	20	0.5610	0.5374	0.5428	0.2368	0.5146	0.5252	0.1780	0.5596	0.5338	0.5412	0.2384	0.5174	0.5246	0.2280
	30	0.5890	0.5670	0.5692	0.2452	0.5370	0.5480	0.1920	0.5840	0.5574	0.5664	0.2490	0.5378	0.5432	0.2140
(100, 1200)	0	0.0616	0.0504	0.0642	0.0700	0.0630	0.0664	0.030	0.0578	0.0460	0.0594	0.0700	0.0608	0.0654	0.0280
	10	0.7400	0.7050	0.7102	0.3280	0.6964	0.7056	0.2680	0.7476	0.6956	0.7214	0.3316	0.7102	0.7152	0.2560
	20	0.8586	0.8406	0.8216	0.3836	0.8108	0.8132	0.3380	0.8532	0.8320	0.8264	0.3876	0.8158	0.8188	0.3300
	30	0.8936	0.8810	0.8582	0.4088	0.8482	0.8534	0.3660	0.8910	0.8830	0.8640	0.4068	0.8520	0.8582	0.3460
Sparse															
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$						$\epsilon_i \sim (3/4)^{1/2}t_8$							
		New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50, 1200)	10	0.2726	0.2470	0.2724	0.1442	0.2570	0.2670	0.1120	0.2858	0.2542	0.2786	0.1562	0.2568	0.2710	0.1360
	20	0.3866	0.3560	0.3768	0.1770	0.3548	0.3678	0.1440	0.3878	0.3588	0.3788	0.1900	0.3574	0.3710	0.1700
	30	0.4478	0.4232	0.4338	0.1952	0.4040	0.4210	0.1620	0.4436	0.4176	0.4350	0.2052	0.4096	0.4218	0.1660
(100, 1200)	10	0.3520	0.2902	0.3554	0.1818	0.3488	0.3584	0.1040	0.3550	0.2942	0.3662	0.1886	0.3532	0.3594	0.0900
	20	0.5502	0.4888	0.5406	0.2490	0.5312	0.5322	0.1820	0.5520	0.4890	0.5412	0.2520	0.5316	0.5408	0.1600
	30	0.6644	0.6110	0.6446	0.2930	0.6286	0.6370	0.2420	0.6758	0.6186	0.6528	0.2962	0.6416	0.6474	0.2360

4 Simulation results

In this section, we conduct numerical simulations to evaluate the finite sample performance of the proposed tests. We compare the proposed test (abbreviated as New) with the ones proposed by Wang and Xu (2021) (abbreviated as WX), Goeman et al. (2006) (abbreviated as GVV), Zhao et al. (2024) (abbreviated as ZLLZ), Zhong and Chen (2011) (abbreviated as ZC), Cui et al. (2018) (abbreviated as CGZ) and Guo et al. (2022) (abbreviated as GZDC). The former two are Bayesian procedures and the latter four are U -statistic-based tests. We also perform simulations to reveal the asymptotic behavior of the T_{BF} under the null hypothesis. The error term ϵ is independently generated from two distributions, a standard normal distribution $N(0, 1)$ or $(3/4)^{1/2}t_8$, where t_8 represents a univariate t -distribution with 8 degrees of freedom. We consider two settings of β_2 : dense or sparse β_2 . For dense β_2 setting, p_n elements in β_2 are nonzero. For sparse β_2 setting, to make simulation settings challenging to testing procedures, we randomly select 25% of the p_n elements in β_2 to be non-zero. The nonzero elements of the two different settings are independently generated from the uniform distribution $U(-c_0, c_0)$, respectively, where c_0 is selected to acquire predetermined value of signal to noise ratio (SNR) which is defined as $SNR = \phi \beta_2^\top \beta_2 \sqrt{\sum_{i=1}^{n-q_n} \lambda_i^2 - (\sum_{i=1}^{n-q_n} \lambda_i)^2 / (n - q_n) / p_n}$. The SNR is set to 0, 10, 20, 30, respectively. $SNR = 0$ is used to examine the type I error rate and the other values are used for local power. We consider two settings for (p_n, n) : first, $p_n/n = 1.6$; and second, $(p_n, n) = (1200, 50)$. The second setting is chosen to approximate the relationship $p_n \approx \exp(n^\alpha)$ with some $\alpha > 0$. For $(p_n, n) = (1200, 50)$, we increase n to 100 for comparison. In all simulations, the elements of $\mathbf{X}_1$ are independently from a standard normal distribution. All the simulation results are based on 5000 independent replications, q_n is fixed to 10, and the nominal significance level of the tests is fixed to be $\alpha = 0.05$. In each case, once $\mathbf{X}_1$ and $\mathbf{X}_2$ are randomly generated, they are held fixed throughout this simulation. We consider the following four scenarios for generating $\mathbf{X}_2$.

Case I. The rows of $\mathbf{X}_2$ are independently generated from $(3/4)^{1/2}t_8(0, \Sigma)$, where $t_8(0, \Sigma)$ represents a multivariate t -distribution with 8 degrees of freedom, mean 0 and covariance structure $\Sigma = (\sigma_{ij})$ with $\sigma_{ij} = 0.5^{|i-j|}$, $i, j = 1, 2, \dots, p_n$.

Case II. The rows of $\mathbf{X}_2$ are independently generated from $N(0, \Sigma)$, where Σ is block diagonal matrix, with block $(p_n/10) \times (p_n/10)$ in which the diagonal entries are 1 and the off-diagonal entries are 0.25.

Case III. The columns of $\mathbf{X}_2$ are independently generated from $N(0, \Sigma)$, where $\Sigma = (\sigma_{ij})$ with $\sigma_{ii} = 1$ and $\sigma_{ij} = 0.1$ if $i \neq j$, for $i, j = 1, \dots, n$.

Case IV. The rows of $\mathbf{X}_2$ are independently generated from $N(0, \Sigma)$, where $\Sigma = (\sigma_{ij})$ with $\sigma_{ii} = 1$ and $\sigma_{ij} = 0.1$ if $i \neq j$, for $i, j = 1, \dots, p_n$.

Case I and II follow Zhao et al. (2024), and case III and IV follow Wang and Xu (2021). The results for cases I–IV with fixed $p_n = 1200$ are summarized in Tables 1, 2, 3 and 4. Due to space limitations, we report only the results for case I with $p_n/n = 1.6$, which are presented in Figs. 1 and 2. It is observed that, when p_n is much larger than n , the proposed test and WX tests maintain the nominal size well across all scenarios, whereas the other methods exhibit slight size distortion in some scenarios.

Table 3 Empirical sizes and powers of T_{BF} , WX , GVV , ZLLZ , ZC , CGZ and GZDC tests at 5% significance level based on 5000 independent replications for case III with $\epsilon_i \sim N(0, 1)$ and $\epsilon_i \sim (3/4)^{1/2}t_8$ under dense and sparse alternatives

Dense											
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$			$\epsilon_i \sim (3/4)^{1/2}t_8$						
		New	WX	GZDC	New	WX	GZDC	CGZ	ZC	ZLLZ	GZDC
(50, 1200)	0	0.0570	0.0476	0.0676	0.1474	0.0540	0.0120	0.0120	0.0540	0.1372	0.0680
	10	0.3634	0.3368	0.3604	0.2792	0.1304	0.0268	0.0268	0.1304	0.2724	0.0560
	20	0.4010	0.3798	0.3946	0.3076	0.1478	0.0344	0.0344	0.1478	0.2970	0.0640
	30	0.4164	0.3980	0.4058	0.3236	0.1580	0.0352	0.0352	0.1580	0.3098	0.0540
(100, 1200)	0	0.0538	0.0460	0.0664	0.0784	0.0472	0.0174	0.0174	0.0472	0.0816	0.0280
	10	0.4876	0.4376	0.4702	0.2552	0.1986	0.0978	0.0978	0.1986	0.2700	0.0360
	20	0.5916	0.5622	0.5362	0.3532	0.2884	0.1496	0.1496	0.2884	0.3618	0.0380
	30	0.6382	0.6152	0.5644	0.4074	0.3414	0.1880	0.1880	0.3414	0.4156	0.0460
Sparse											
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$			$\epsilon_i \sim (3/4)^{1/2}t_8$						
		New	WX	GZDC	New	WX	GZDC	CGZ	ZC	ZLLZ	GZDC
(50, 1200)	10	0.2476	0.2086	0.2571	0.2010	0.0812	0.0174	0.0174	0.0812	0.2004	0.0460
	20	0.3176	0.2850	0.3225	0.2360	0.1024	0.0206	0.0206	0.1024	0.2326	0.0540
	30	0.3536	0.3242	0.3572	0.2614	0.1168	0.0248	0.0248	0.1168	0.2540	0.0640
	10	0.2694	0.2026	0.2976	0.1362	0.0912	0.0420	0.0420	0.0912	0.1354	0.0220
(100, 1200)	20	0.3838	0.3104	0.3936	0.1896	0.1320	0.0628	0.0628	0.1320	0.1852	0.0360
	30	0.4608	0.3890	0.4464	0.2368	0.1746	0.0850	0.0850	0.1746	0.2310	0.0500

Table 4 Empirical sizes and powers of T_{BP} , WX , GVV , $ZLLZ$, ZC , CGZ and $GZDC$ tests at 5% significance level based on 5000 independent replications for case IV with $\epsilon_i \sim N(0, 1)$ and $\epsilon_i \sim (3/4)^{1/2}t_8$ under dense and sparse alternatives

Dense															
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$						$\epsilon_i \sim (3/4)^{1/2}t_8$							
		New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50,1200)	0	0.0666	0.0540	0.0698	0.0820	0.0724	0.0782	0.0600	0.0560	0.0486	0.0642	0.0732	0.0630	0.0668	0.0640
	10	0.3940	0.3650	0.3884	0.3640	0.3608	0.3664	0.1980	0.3908	0.3624	0.3880	0.3632	0.3658	0.3732	0.1940
	20	0.4356	0.4144	0.4284	0.3940	0.3968	0.4026	0.2300	0.4312	0.4116	0.4218	0.3958	0.4010	0.4066	0.2120
	30	0.4494	0.4296	0.4382	0.4064	0.4138	0.4160	0.2360	0.4494	0.4300	0.4370	0.4106	0.4168	0.4194	0.2300
(100,1200)	0	0.0554	0.0428	0.0668	0.0716	0.0660	0.0668	0.0400	0.0536	0.0458	0.0680	0.0690	0.0650	0.0676	0.0480
	10	0.5058	0.4496	0.4832	0.4746	0.4802	0.4832	0.3120	0.4998	0.4492	0.4846	0.4736	0.4800	0.4854	0.3180
	20	0.6030	0.5672	0.5482	0.5322	0.5440	0.5468	0.3740	0.6016	0.5720	0.5526	0.5322	0.5474	0.5492	0.3860
	30	0.6456	0.6290	0.5772	0.5590	0.5720	0.5772	0.4080	0.6516	0.6280	0.5812	0.5604	0.5790	0.5824	0.3960
Sparse															
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$						$\epsilon_i \sim (3/4)^{1/2}t_8$							
		New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50,1200)	10	0.2436	0.2122	0.2540	0.2438	0.2384	0.2432	0.1460	0.2506	0.2140	0.2570	0.2464	0.2398	0.2458	0.1460
	20	0.3132	0.2832	0.3190	0.2998	0.2980	0.3042	0.1700	0.3158	0.2852	0.3246	0.3026	0.3008	0.3042	0.1520
	30	0.3502	0.3202	0.3508	0.3280	0.3268	0.3336	0.2040	0.3542	0.3276	0.3564	0.3320	0.3330	0.3352	0.1800
(100,1200)	10	0.2678	0.1864	0.2870	0.2804	0.2810	0.2864	0.1380	0.2788	0.1872	0.2912	0.2886	0.2844	0.2874	0.1420
	20	0.3610	0.2922	0.3840	0.3762	0.3794	0.3822	0.1940	0.3624	0.2938	0.3832	0.3822	0.3822	0.3850	0.1880
	30	0.4344	0.3622	0.4370	0.4226	0.4302	0.4348	0.2380	0.4378	0.3672	0.4342	0.4292	0.4302	0.4360	0.2380

Table 5 Sensitivity analysis of the prior hyperparameters for the T_{BF} test at 5% significance level

Dense											
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$			$\epsilon_i \sim (3/4)^{1/2} t_8$						
		p^2	n^2	n	$1/n$	$1/p$	$1/n^2$	p^2	n^2	n	$1/p$
(50,1200)	0	0.0554	0.0544	0.0488	0.0476	0.5392	0.3568	0.0704	0.0690	0.0636	0.5352
	10	0.4988	0.5172	0.4988	0.4944	0.5198	0.3218	0.4964	0.5156	0.4972	0.5274
	20	0.5806	0.6034	0.6000	0.5934	0.5196	0.3164	0.5748	0.5992	0.5942	0.5250
	30	0.6086	0.6378	0.6428	0.6360	0.5208	0.3128	0.6058	0.6346	0.6372	0.5260
(100,1200)	0	0.0522	0.0504	0.0420	0.0558	0.4288	0.4064	0.0786	0.0784	0.0708	0.4210
	10	0.7486	0.7652	0.7656	0.7682	0.4146	0.3996	0.7478	0.7660	0.7538	0.4320
	20	0.8608	0.8774	0.9022	0.9034	0.4218	0.4014	0.8676	0.8792	0.8942	0.4286
	30	0.8972	0.9144	0.9404	0.9354	0.4218	0.3954	0.9000	0.9128	0.9354	0.4228
Sparse											
(n, p)	SNR	$\epsilon_i \sim N(0, 1)$			$\epsilon_i \sim (3/4)^{1/2} t_8$						
		p^2	n^2	n	$1/n$	$1/p$	$1/n^2$	p^2	n^2	n	$1/p$
(50,1200)	10	0.2540	0.2614	0.2266	0.2210	0.5128	0.3374	0.2740	0.2748	0.2504	0.5256
	20	0.3844	0.3968	0.3650	0.3586	0.5118	0.3296	0.3894	0.3950	0.3750	0.5282
	30	0.4510	0.4676	0.4478	0.4406	0.5104	0.3274	0.4528	0.4682	0.4510	0.5262
	10	0.3254	0.3258	0.2720	0.2890	0.4132	0.4132	0.3510	0.3528	0.3042	0.4348
(100,1200)	20	0.5380	0.5522	0.5052	0.5134	0.4148	0.4054	0.5540	0.5610	0.5282	0.4260
	30	0.6610	0.6770	0.6542	0.6592	0.4130	0.4076	0.6656	0.6820	0.6684	0.4248

Results are based on 5000 independent replications for case I, where $\epsilon_i \sim N(0, 1)$ and $\epsilon_i \sim (3/4)^{1/2} t_8$

Table 6 The running time (seconds) of T_{BF} , WX, GVV, ZLLZ, ZC, CGZ and GZDC tests for case I with $\epsilon_i \sim N(0, 1)$ and $SNR \in \{0, 10, 20, 30\}$ under dense alternatives

(n, p_n)	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
(50,80)	53.19	52.21	52.32	330.92	549.38	354.89	2759.40
(50,1200)	126.55	125.86	125.77	2967.79	4567.85	1383.03	3196.74
(50,5000)	788.64	784.90	796.32	13202.16	19705.09	6994.50	4844.97

The phenomenon is mitigated as the sample size increases to 100. For the $p_n/n = 1.6$ setting, however, all tests can control the nominal size well, although the WX test is slightly conservative. In terms of empirical power, under the $p_n \approx \exp(n^\alpha)$, $\alpha > 0$ setting, the proposed test substantially outperforms all competing methods in all dense scenarios and most sparse scenarios with an SNR of 30. It exhibits slightly lower power than the GVV test in some sparse scenarios with $SNR = 10$ and 20. In the $p_n/n = 1.6$ setting, the proposed test is a top performer, except in sparse scenarios with an $SNR = 10$, where it is slightly outperformed by GVV. However, in both settings where our test is not the top performer, the difference is quite small. These findings are reasonable because the proposed test includes a ratio of quadratic forms, which yields higher power when the signal is dense, whereas the GVV and CGZ tests are more adept at detecting sparse alternatives. For $n = 50$, the tests fail to achieve power approximating 1 even with $SNR = 30$. This observation is consistent with the fact that no test can guarantee reasonable power, even when the true model deviates substantially from the null hypothesis under “large p , small q ” setting. Overall, our test still shows remarkable performance across all scenarios. As the sample size increases, the empirical power of all tests rises, as expected. In addition, our new test is relatively robust against non-normal error distributions, which broadens its applicability, even though normality is assumed in our paper. Therefore, the proposed test is recommended for testing the overall significance of high-dimensional regression coefficients in linear models due to its attractive performance in terms of both empirical size and power.

Furthermore, we use QQ-plots presented in Figs. 3, 4 and 5 to examine the accuracy of the limiting null distribution of our test statistic T_{BF} with different sample sizes n . The points in Figs. 3, 4 and 5 align closely with the identity line across all considered scenarios (case I–IV), indicating that the finite-sample distribution of T_{BF} is well approximated by $N(0, 1)$. This strong agreement holds for errors drawn from both the standard normal distribution and the centered t -distribution. These findings provide empirical validations for the theoretical results established in Theorem 1 (Tables 5, 6).

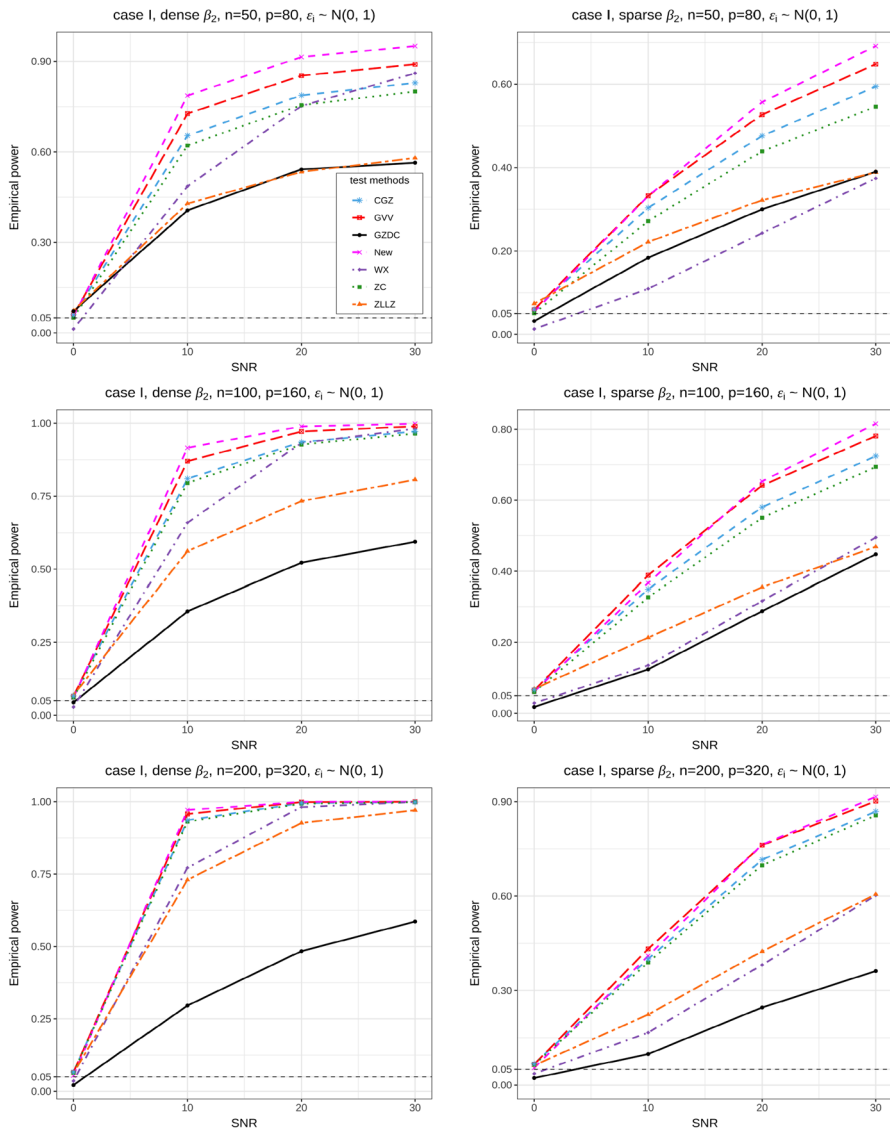


Fig. 1 Empirical sizes and powers of T_{BF} test at 5% significance level for case I with $\epsilon_i \sim N(0, 1)$ under dense and sparse alternatives across different (n, p) combinations

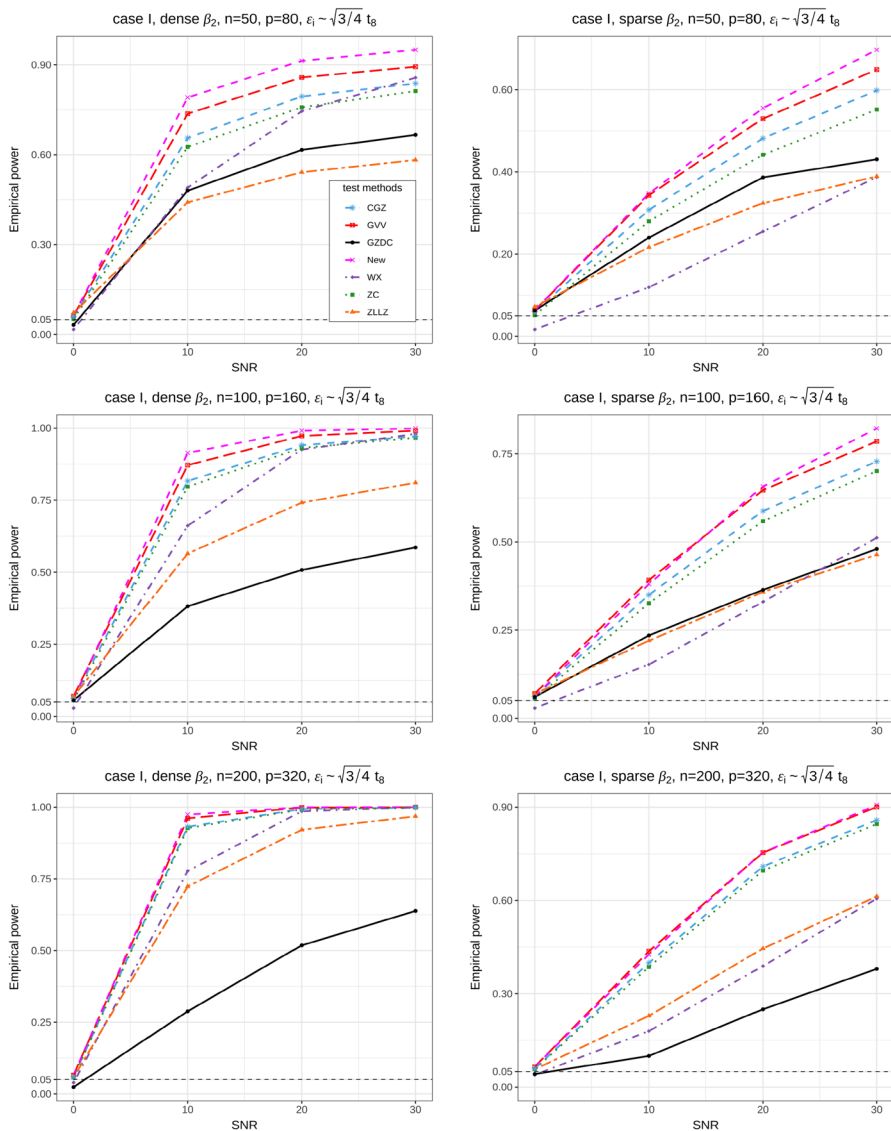


Fig. 2 Empirical sizes and powers of T_{BF} test at 5% significance level for case I with $\epsilon_i \sim (3/4)^{1/2} t_8$ under dense and sparse alternatives across different (n, p) combinations

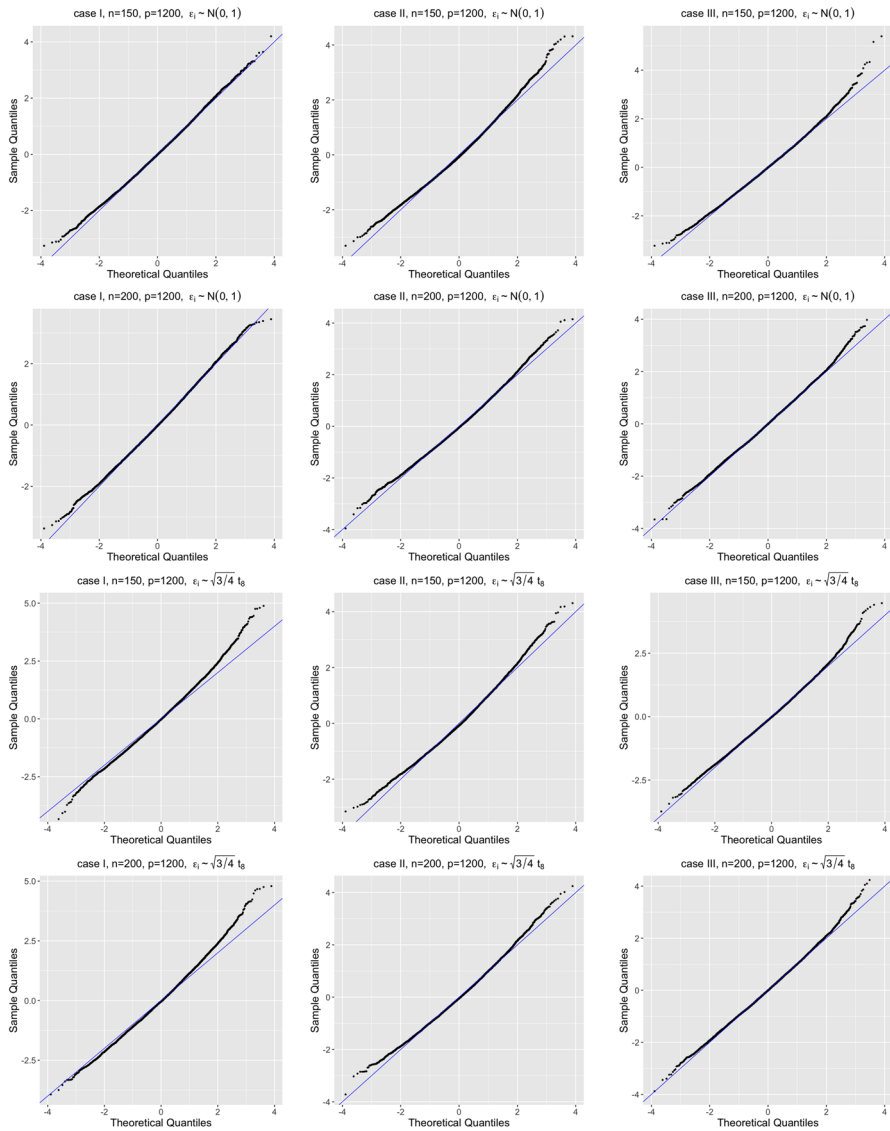


Fig. 3 QQ-plots of T_{BF} under the null hypothesis for cases I (the first column), case II (the second column) and case III (the third column) with varying n and fixed p_n for both $\epsilon_i \sim N(0, 1)$ and $\epsilon_i \sim (3/4)^{1/2} t_8$

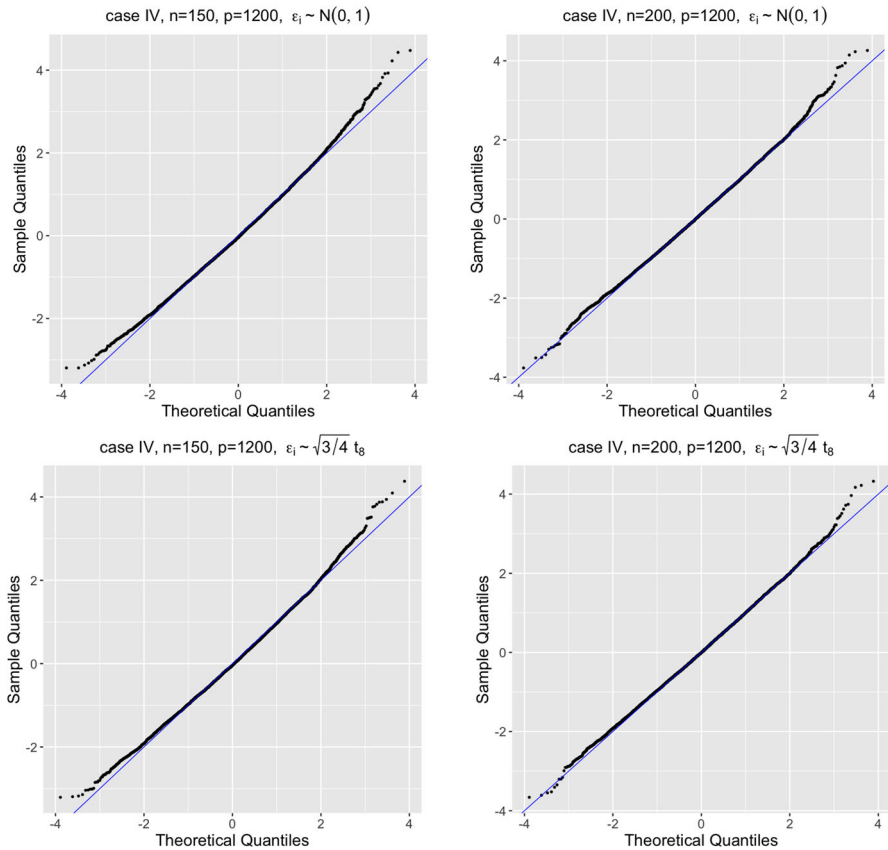


Fig. 4 QQ-plots of T_{BF} under the null hypothesis for case IV with $\epsilon_i \sim N(0, 1)$ (the first row) and $\epsilon_i \sim (3/4)^{1/2} t_8$ (the second row), for different n and fixed $p = 1200$

5 Applications

In this section, we apply the proposed test to two real data examples to further investigate its practical utility.

5.1 Example 1

We analyze a real-world dataset concerning the production rate of riboflavin (Vitamin B2) in *Bacillus subtilis*. This dataset was previously analyzed by (Van de Geer et al. 2014) and is publicly available in the R package *hdi*. The response variable, y , is the logarithm of the riboflavin production rate for $n = 71$ samples of genetically engineered mutants of *bacillus subtilis*. The model includes $p_n = 4088$ covariates, which are the log-transformed expression levels of 4088 genes. We model the data using a high-dimensional linear model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$. First, we consider a global hypothesis test for the overall significance of the covariates: $\mathcal{H}_0 : \boldsymbol{\beta} = 0$ v.s. $\mathcal{H}_1 :$

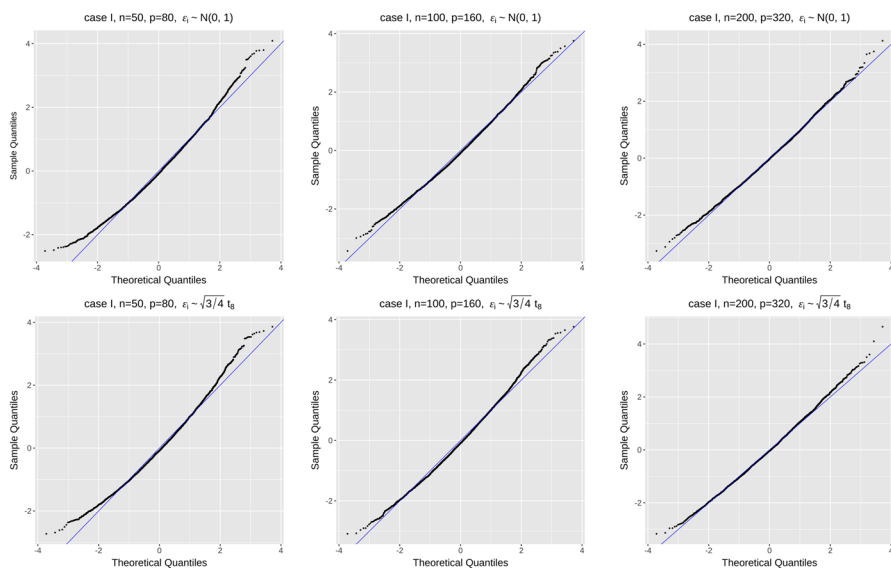


Fig. 5 QQ-plots of T_{BF} under the null hypothesis for case I with $\epsilon_i \sim N(0, 1)$ (the first row) and $\epsilon_i \sim (3/4)^{1/2} t_8$ (the second row), for different (n, p) combinations

$\beta \neq 0$. The p values from our proposed test and five competing methods (WX, GVV, ZLLZ, ZC, CGZ and GZDC) are presented in Table 7. At a significance level of 0.05, all tests except for CGZ and GZDC tests reject the null hypothesis. Our proposed test yields a p value of 0. This indicates a serious deviation from the null hypothesis, meaning that the gene expression levels are significant for predicting the riboflavin production rate.

Next, we investigate the significance of a subset of genes identified via variable selection. We apply the Broken adaptive ridge (Dai et al. 2018, BAR) method to the 4088 predictors to select important variables for the response. The BAR method identifies a set of 56 active predictors, whose coefficients correspond to β_1 in the linear model (1). Testing whether coefficients are exactly zero to avoid missing important variables is crucial. Hence, our interest lies in testing the significance of a large number of remaining variables, whose coefficients were shrunk to zero by BAR. Table 7 reports the p values for this partial testing problem from all considered methods. The New, WX, GVV, CGZ tests yield p values larger than 0.05, suggesting that the null hypothesis $\mathcal{H}_0 : \beta_2 = 0$, cannot be rejected. This conclusion is consistent with the variable selection results from the BAR method.

5.2 Example 2

To gain a broad perspective of gene regulation in the mammalian eye and identify genetic variants relevant to human eye disease, Scheetz et al. (2006) performed an expression quantitative trait locus mapping study in the laboratory rats. The dataset (GEO series GSE5680) can be found at <https://www.ncbi.nlm.nih.gov/geo/query/acc>.

Table 7 The p values of the proposed tests and the WX, GVV, ZLLZ, ZC, CGZ and GZDC tests for example 1

Tests	$\mathcal{H}_0 : \beta = 0 \text{ v.s. } \mathcal{H}_1 : \beta \neq 0.$						$\mathcal{H}_0 : \beta_2 = 0 \text{ v.s. } \mathcal{H}_1 : \beta_2 \neq 0.$							
	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
p value	0	1.59×10^{-12}	0	0.0041	5.01×10^{-5}	0.4848	0.4847	0.0593	0.9663	0.0689	0.0092	0.0002	0.4786	NA

“NA” denotes singularity errors occurring when running the GZDC procedure

[cgi?acc=GSE5680](#) (Scheetz et al. 2006). The data processing procedure, detailed in the Supplementary material, resulted in a final set of 18,976 probes that were considered to be reliably expressed and sufficiently variable. Scheetz et al. (2006) identified 22 of 18,976 gene probes as important in regulating mammalian eye gene expression. We consider a linear regression model for the response gene TRIM32, which relates to the retinal disease called Bardet–Biedl syndrome. From the 22 significant probes identified by Scheetz et al. (2006), we randomly select a subset of 16 to serve as known active predictors for the TRIM32 gene. We then test whether the remaining 18,959 genes have a significant effect on the response. The p values for our proposed test and six competing methods (WX, GVV, ZLLZ, ZC, CGZ and GZDC) for this partial hypothesis testing are presented in Table 8. For comparison, we also report in the same table the p values for the global significance test of all regression coefficients. From Table 8, it can be seen that our proposed method (New) and two competing tests (GVV, ZC) correctly reject the null hypothesis for both the global and partial hypotheses testing. This outcome is expected, as the sets of variables under consideration for these tests contain at least 22 and 6 previously identified significant genes, respectively.

6 Conclusion and discussion

In this paper, we propose a novel method for testing the overall significance of a large number of remaining predictors, given that a few predictors are known to have a significant effect on the response in high-dimensional linear models with fixed design. Our test is based on Bayes factor. We construct a unified Bayes factor-based testing framework for the overall significance of regression coefficients that is applicable across all relationships between the parameter dimension p_n and the sample size n . We consider three distinct settings: (i) $p_n/(n - q_n) \rightarrow c \in (1, \infty]$, (ii) $p_n/(n - q_n) \rightarrow c \in (0, 1)$, and (iii) $p_n/(n - q_n) \rightarrow 0$ while $p_n \rightarrow \infty$. For the high-dimensional setting (i), we establish the asymptotic normality of the proposed test statistic, T_{BF} , under both the null hypothesis and local alternatives. The asymptotic power function of T_{BF} is derived and an analysis of it against the existing methods is also provided. Notably, our theoretical results are applicable to the setting (ii), with the main difference being that the dimensionality of key matrices (e.g., $\tilde{\Gamma}$, $\tilde{\Lambda}_2$) is determined by p_n rather than $n - q_n$. For setting (iii), under some certain conditions, the expression of the proposed test statistic is further simplified. The corresponding asymptotic results for this simplified test is established in the Supplementary material. Extensive simulation results demonstrate that our test outperforms several competing methods. It also exhibits robustness against non-normal error distributions. Furthermore, the QQ-plots confirm that the asymptotic null distribution of T_{BF} closely follows the standard normal distribution. We applied the proposed test to two real high-dimensional datasets to illustrate its practical effectiveness, and the analysis yields promising results. Therefore, we recommend the Bayes factor as a test statistic for testing the regression coefficients in high dimensional linear models. A direction worthy of consideration for future work would

Table 8 The p values of the proposed tests and the WX, GVV, ZLLZ, ZC, CGZ and GZDC tests for example 2

Tests	$\mathcal{H}_0 : \beta = 0$ v.s. $\mathcal{H}_1 : \beta \neq 0.$						$\mathcal{H}_0 : \beta_2 = 0$ v.s. $\mathcal{H}_1 : \beta_2 \neq 0.$							
	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC	New	WX	GVV	ZLLZ	ZC	CGZ	GZDC
p value	0	9.80×10^{-39}	0	0.0962	0	0.4257	0.4258	2.07×10^{-21}	1	6.62×10^{-20}	0.0960	0	0.4288	NA

“NA” denotes singularity errors occurring when running the GZDC procedure

be to extend the Bayes factor approach to hypothesis testing in other high-dimensional models.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00362-026-01884-1>.

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Appendix

Proof of Lemma 1 Let $\mathbf{B} = (n - q_n)\tilde{\mathbf{U}}_2\tilde{\mathbf{\Gamma}}\tilde{\mathbf{U}}_2^\top - \zeta_x\mathbf{I}_{n-q_n}$. It is straightforward to show that

$$\begin{aligned}\lambda_1(\mathbf{B}^2) &= \max_{1 \leq i \leq n-q_n} \left((n - q_n) \frac{\lambda_i}{\gamma_i + \lambda_i} - \text{tr}(\tilde{\mathbf{\Gamma}}) - \sqrt{\sigma_0^2 x} \right)^2, \\ \text{tr}(\mathbf{B}^2) &= \left((n - q_n)^2/2 + (n - q_n)x^2 \right) \sigma_0^2,\end{aligned}\quad (\text{A.1})$$

then

$$\begin{aligned}\frac{\lambda_1(\mathbf{B}^2)}{\text{tr}(\mathbf{B}^2)} &\leq \frac{2 \max_{1 \leq i \leq n-q_n} \left((n - q_n) \frac{\lambda_i}{\gamma_i + \lambda_i} - \text{tr}(\tilde{\mathbf{\Gamma}}) \right)^2}{((n - q_n)^2/2 + (n - q_n)x^2) \sigma_0^2} + \frac{2\sigma_0^2 x^2}{((n - q_n)^2/2 + (n - q_n)x^2) \sigma_0^2} \\ &\leq \frac{2 \max_{1 \leq i \leq n-q_n} \left((n - q_n) \frac{\lambda_i}{\gamma_i + \lambda_i} - \text{tr}(\tilde{\mathbf{\Gamma}}) \right)^2}{(n - q_n)^2 \left(\text{tr}(\tilde{\mathbf{\Gamma}}^2) - \frac{(\text{tr}(\tilde{\mathbf{\Gamma}}))^2}{n - q_n} \right) + 2(n - q_n)^2 x^2 \left(\text{tr}(\tilde{\mathbf{\Gamma}}^2) - \frac{(\text{tr}(\tilde{\mathbf{\Gamma}}))^2}{n - q_n} \right)} + \frac{2}{n - q_n} \\ &\rightarrow 0,\end{aligned}\quad (\text{A.2})$$

where the last step holds by the fact that $\text{tr}(\tilde{\mathbf{\Gamma}}^2) - (\text{tr}(\tilde{\mathbf{\Gamma}})/(n - q_n))^2 > 0$ and condition (14). Thus, the necessary condition of Proposition 2 of (Wang and Xu 2021) is satisfied. Under the alternative hypotheses, since $\tilde{\mathbf{y}}^\top \mathbf{B} \tilde{\mathbf{y}} = \boldsymbol{\epsilon}^\top \mathbf{U}_1 \mathbf{B} \mathbf{U}_1^\top \boldsymbol{\epsilon} + 2\boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \mathbf{U}_1^\top \boldsymbol{\epsilon} + \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2$, by means of the Proposition 2 of (Wang and Xu 2021), we have

$$\begin{aligned}\Pr \left((n - q_n) \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \tilde{\mathbf{y}}}{\tilde{\mathbf{y}}^\top \tilde{\mathbf{y}}} \leq \zeta_x \right) &= \Pr(\tilde{\mathbf{y}}^\top \mathbf{B} \tilde{\mathbf{y}} \leq 0) \\ &= \Pr(\boldsymbol{\epsilon}^\top \mathbf{U}_1 \mathbf{B} \mathbf{U}_1^\top \boldsymbol{\epsilon} + 2\boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \mathbf{U}_1^\top \boldsymbol{\epsilon} + \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2 > 0) \\ &= \Pr \left(\frac{\boldsymbol{\epsilon}^\top \mathbf{U}_1 \mathbf{B} \mathbf{U}_1^\top \boldsymbol{\epsilon} + 2\boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \mathbf{U}_1^\top \boldsymbol{\epsilon} - \phi^{-1} \text{tr}(\mathbf{B})}{\sqrt{2\phi^{-2} \text{tr}(\mathbf{B}^2) + 4\phi^{-1} \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B}^2 \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2}} \leq \frac{-\boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2 - \phi^{-1} \text{tr}(\mathbf{B})}{\sqrt{2\phi^{-2} \text{tr}(\mathbf{B}^2) + 4\phi^{-1} \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B}^2 \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2}} \right) \\ &= \Phi \left(\frac{-\boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2 - \phi^{-1} \text{tr}(\mathbf{B})}{\sqrt{2\phi^{-2} \text{tr}(\mathbf{B}^2) + 4\phi^{-1} \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B}^2 \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2}} \right) + o(1),\end{aligned}\quad (\text{A.3})$$

Additionally, by some elementary calculation,

$$\begin{aligned}\operatorname{tr}(\mathbf{B}) &= -(n - q_n)\sqrt{\sigma_0^2}x, \\ \operatorname{tr}(\mathbf{B}^2) &= (1 + o(1))(n - q_n)^2\sigma_0^2/2, \\ \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B} \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2 &= \boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left((n - q_n) \tilde{\Gamma} - \zeta_x \mathbf{I}_{n-q_n} \right) \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2, \\ \boldsymbol{\beta}_2^\top \tilde{\mathbf{X}}_2^\top \mathbf{B}^2 \tilde{\mathbf{X}}_2 \boldsymbol{\beta}_2 &= \boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left((n - q_n) \tilde{\Gamma} - \zeta_x \mathbf{I}_{n-q_n} \right)^2 \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2.\end{aligned}\tag{A.4}$$

Substituting (A.4) into (A.3) gives the desired results. $\square$

Proof of Proposition 1 Fix an $x \in \mathbb{R}$, by the local alternatives in (16),

$$\begin{aligned}& \boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\frac{\tilde{\Gamma} - \frac{\zeta_x}{n-q_n} \mathbf{I}_{n-q_n}}{\sqrt{\sigma_0^2}} \right)^2 \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2 \\ & \leq \frac{2\boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\tilde{\Gamma} - \frac{\operatorname{tr}(\tilde{\Gamma})}{n-q_n} \mathbf{I}_{n-q_n} \right)^2 \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2}{\sigma_0^2} + \frac{2\boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\frac{\sqrt{\sigma_0^2}}{n-q_n} x \mathbf{I}_{n-q_n} \right)^2 \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2}{\sigma_0^2} \\ & \leq \max_{1 \leq i \leq n-q_n} (\lambda_i \omega_i^2) \frac{\sigma_0^2}{\sigma_0^2} + \frac{2\boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2}{(n - q_n)^2} \\ & = o(1) + O\left((n - q_n)^{-3/2}\right) \rightarrow 0\end{aligned}\tag{A.5}$$

and $\phi \boldsymbol{\beta}_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \tilde{\mathbf{V}}_2^\top \boldsymbol{\beta}_2 / (n - q_n) = O((n - q_n)^{-1/2}) \rightarrow 0$. This, together with (A.5) and (15) gives the desired result. $\square$

Proof of Theorem 1 It is straightforward to show that

$$\begin{aligned}& \frac{2 \log \text{BF}_{10} - \sum_{i=1}^{n-q_n} \log \gamma_i + \sum_{i=1}^{n-q_n} \log(\gamma_i + \lambda_i) + (n - q_n) \left(\log(1 - \mu_0) - \frac{\operatorname{Var}_0(T)}{2(1 - \mu_0)^2} \right)}{\sqrt{\sigma_0^2}/(1 - \mu_0)} \\ & = \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} \frac{1 - 2\mu_0}{1 - \mu_0} + \frac{(n - q_n)(T^2 - \mu_0^2)}{2\mu_0 \sqrt{\sigma_0^2}} \frac{\mu_0}{1 - \mu_0} + o_p(1).\end{aligned}\tag{A.6}$$

Let

$$W = \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} \frac{1 - 2\mu_0}{1 - \mu_0} + \frac{(n - q_n)(T^2 - \mu_0^2)}{2\mu_0 \sqrt{\sigma_0^2}} \frac{\mu_0}{1 - \mu_0}.$$

We first prove (i). Under the null hypotheses,

$$T = \frac{\boldsymbol{\epsilon}^\top \mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top \boldsymbol{\epsilon}}{\boldsymbol{\epsilon}^\top \mathbf{U}_1 \mathbf{U}_1^\top \boldsymbol{\epsilon}}.$$

Substituting the above equation and $\beta_2 = 0$ into (17) gives

$$\Pr \left((n - q_n) \frac{\boldsymbol{\epsilon}^\top \mathbf{U}_1 \tilde{\mathbf{U}}_2 \tilde{\mathbf{\Gamma}} \tilde{\mathbf{U}}_2^\top \mathbf{U}_1^\top \boldsymbol{\epsilon}}{\boldsymbol{\epsilon}^\top \mathbf{U}_1 \mathbf{U}_1^\top \boldsymbol{\epsilon}} \leq \text{tr}(\tilde{\mathbf{\Gamma}}) + \sqrt{\sigma_0^2} x \right) = \Phi(x) + o(1), \quad (\text{A.7})$$

which means that $(n - q_n)(T - \mu_0)/\sqrt{\sigma_0^2} \xrightarrow{d} N(0, 1)$. Next we turn to $\frac{(n - q_n)(T^2 - \mu_{02})}{2\mu_0\sqrt{\sigma_0^2}}$.

Since

$$T^2 - \mu_{02} = 2\mu_0(T - \mu_0) + \left[(T - \mu_0)^2 - \text{Var}_0(T) \right],$$

then

$$\begin{aligned} \frac{(n - q_n)(T^2 - \mu_{02})}{2\mu_0\sqrt{\sigma_0^2}} &= \frac{(n - q_n)(2\mu_0(T - \mu_0) + [(T - \mu_0)^2 - \text{Var}_0(T)])}{2\mu_0\sqrt{\sigma_0^2}} \\ &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} + \frac{(n - q_n)[(T - \mu_0)^2 - \text{Var}_0(T)]}{2\mu_0\sqrt{\sigma_0^2}}. \end{aligned} \quad (\text{A.8})$$

Let $R = \frac{(n - q_n)[(T - \mu_0)^2 - \text{Var}_0(T)]}{2\mu_0\sqrt{\sigma_0^2}}$, $E(R) = 0$. With the aid of moment generating functions, by Bao and Kan (2013) and some tedious calculations,

$$\begin{aligned} E_0[T^3] &= \frac{(\text{tr}(\tilde{\mathbf{\Gamma}}))^3 + 6\text{tr}(\tilde{\mathbf{\Gamma}})\text{tr}(\tilde{\mathbf{\Gamma}}^2) + 8\text{tr}(\tilde{\mathbf{\Gamma}}^3)}{(n - q_n)(n - q_n + 2)(n - q_n + 4)}, \\ E_0[T^4] &= \frac{(\text{tr}(\tilde{\mathbf{\Gamma}}))^4 + 12(\text{tr}(\tilde{\mathbf{\Gamma}}))^2\text{tr}(\tilde{\mathbf{\Gamma}}^2) + 32\text{tr}(\tilde{\mathbf{\Gamma}})\text{tr}(\tilde{\mathbf{\Gamma}}^3) + 12(\text{tr}(\tilde{\mathbf{\Gamma}}^2))^2 + 48\text{tr}(\tilde{\mathbf{\Gamma}}^4)}{(n - q_n)(n - q_n + 2)(n - q_n + 4)(n - q_n + 6)}. \end{aligned}$$

Hence

$$\begin{aligned} E_0[(T - \mu_0)^4] &= E_0(T^4) - 4\mu_0 E_0(T^3) + 6\mu_0^2 E_0(T^2) - 4\mu_0^3 E_0(T) + \mu_0^4 \\ &= \frac{1}{(n - q_n)^4(n - q_n + 2)(n - q_n + 4)(n - q_n + 6)} \\ &\quad \times \left(12(n - q_n - 12)(\text{tr}(\tilde{\mathbf{\Gamma}}))^4 - 24(n - q_n)(n - q_n - 12)(\text{tr}(\tilde{\mathbf{\Gamma}}))^2\text{tr}(\tilde{\mathbf{\Gamma}}^2) \right. \\ &\quad \left. - 192(n - q_n)^2\text{tr}(\tilde{\mathbf{\Gamma}})\text{tr}(\tilde{\mathbf{\Gamma}}^3) + 12(n - q_n)^3(\text{tr}(\tilde{\mathbf{\Gamma}}^2))^2 + 48(n - q_n)^3\text{tr}(\tilde{\mathbf{\Gamma}}^4) \right) \\ &= O((n - q_n)^{-2}). \end{aligned} \quad (\text{A.9})$$

Combining (A.9) and the order of $\text{Var}_0(T)$ implies that

$$\text{Var}(R) = \frac{(n - q_n)^2 (\mathbb{E}_0[(T - \mu_0)^4] - (\text{Var}_0(T))^2)}{4\mu_0^2\sigma_0^2} = O((n - q_n)^{-1}).$$

This, combined with the fact that $\mathbb{E}(R) = 0$ yields $R = o_p(1)$. Hence, according to (A.8),

$$\begin{aligned} \frac{(n - q_n)(T^2 - \mu_{02})}{2\mu_0\sqrt{\sigma_0^2}} &= \frac{(n - q_n)(2\mu_0(T - \mu_0) + [(T - \mu_0)^2 - \text{Var}_0(T)])}{2\mu_0\sqrt{\sigma_0^2}} \\ &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} + o_p(1). \end{aligned} \quad (\text{A.10})$$

Therefore

$$\begin{aligned} W &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} \frac{1 - 2\mu_0}{1 - \mu_0} + \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} \frac{\mu_0}{1 - \mu_0} + o_p(1) \\ &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} + o_p(1). \end{aligned} \quad (\text{A.11})$$

By now, we can conclude that $W \xrightarrow{d} N(0, 1)$. This completes the proof of (i).

We next prove (ii). Under the local alternative hypotheses in (16), the analysis involves calculations with ratios of noncentral quadratic forms. By applying Taylor expansion and following Bao and Kan (2013), after some tedious and complicated calculations, we obtain

$$\begin{aligned} \mathbb{E}_1(T) &= \frac{\mu_A}{\mu_B} - \frac{\text{Cov}(Q_A, Q_B)}{\mu_B^2} + \frac{\mu_A}{\mu_B^3} \text{Var}(Q_B) + O((n - q_n)^{-3/2}), \\ \mathbb{E}_1(T^2) &= \frac{\mu_A^2}{\mu_B^2} + \frac{\text{Var}(Q_A)}{\mu_B^2} + \frac{3\mu_A^2}{\mu_B^4} \text{Var}(Q_B) - \frac{4\mu_A}{\mu_B^3} \text{Cov}(Q_A, Q_B) + O((n - q_n)^{-3/2}), \\ \text{Var}_1(T) &= \frac{\text{Var}(Q_A)}{\mu_B^2} - \frac{\mu_A^2}{\mu_B^4} \text{Var}(Q_B) + O((n - q_n)^{-3/2}), \\ \text{Var}_1(T^2) &= \frac{4\mu_A^2}{\mu_B^4} \text{Var}(Q_A) \\ &\quad + \frac{4\mu_A^4}{\mu_B^6} \text{Var}(Q_B) - \frac{8\mu_A^3}{\mu_B^5} \text{Cov}(Q_A, Q_B) + O((n - q_n)^{-3/2}), \\ \mathbb{E}_1(T^3) &= \frac{\mu_A^3}{\mu_B^3} + \frac{3\mu_A}{\mu_B^3} \text{Var}(Q_A) \\ &\quad + \frac{6\mu_A^3}{\mu_B^5} \text{Var}(Q_B) - \frac{9\mu_A^2}{\mu_B^4} \text{Cov}(Q_A, Q_B) + O((n - q_n)^{-3/2}), \\ \mathbb{E}_1(T^4) &= \frac{\mu_A^4}{\mu_B^4} + \frac{6\mu_A^2}{\mu_B^4} \text{Var}(Q_A) \end{aligned}$$

$$+ \frac{10\mu_A^4}{\mu_B^6} \text{Var}(Q_B) - \frac{16\mu_A^3}{\mu_B^5} \text{Cov}(Q_A, Q_B) + O((n - q_n)^{-3/2}), \quad (\text{A.12})$$

where

$$\begin{aligned} \xi_A &= \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \tilde{\Gamma} \tilde{\mathbf{V}}_2^\top \beta_2, \\ \xi_B &= \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \tilde{\mathbf{V}}_2^\top \beta_2, \\ \mu_A &= \phi^{-1} \text{tr}(\tilde{\Gamma}) + \xi_A, \\ \mu_B &= (n - q_n) \phi^{-1} + \xi_B, \\ \text{Var}(Q_A) &= 2\phi^{-2} \text{tr}(\tilde{\Gamma}^2) + 4\phi^{-1} \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \tilde{\Gamma}^2 \tilde{\mathbf{V}}_2^\top \beta_2, \\ \text{Var}(Q_B) &= 2(n - q_n) \phi^{-2} + 4\phi^{-1} \beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\Lambda}_2^2 \tilde{\mathbf{V}}_2^\top \beta_2, \\ \text{Cov}(Q_A, Q_B) &= 2\phi^{-2} \text{tr}(\tilde{\Gamma}) + 4\phi^{-1} \xi_A. \end{aligned} \quad (\text{A.13})$$

Hence, the order of $E_1[(T - \mu_0)^2]$ and $E_1[(T - \mu_0)^4]$ can be calculated as

$$\begin{aligned} E_1[(T - \mu_0)^2] &= O((n - q_n)^{-1}), \\ E_1[(T - \mu_0)^4] &= O((n - q_n)^{-3/2}). \end{aligned} \quad (\text{A.14})$$

Combining (A.14) and the order of $\text{Var}_0(T)$, μ_0 , we have

$$\text{Var}(R) = \frac{(n - q_n)^2 (E_1[(T - \mu_0)^4] - (E_1(T - \mu_0)^2)^2)}{4\mu_0^2 \sigma_0^2} = O((n - q_n)^{-1/2}).$$

This, combined with the fact that $E(R) = O((n - q_n)^{-1/2})$ under the local alternatives, yields $R = o_p(1)$. Hence, according to (A.8),

$$\begin{aligned} \frac{(n - q_n)(T^2 - \mu_{02})}{2\mu_0 \sqrt{\sigma_0^2}} &= \frac{(n - q_n) (2\mu_0(T - \mu_0) + [(T - \mu_0)^2 - \text{Var}_0(T)])}{2\mu_0 \sqrt{\sigma_0^2}} \\ &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} + o_p(1). \end{aligned} \quad (\text{A.15})$$

Therefore under the local alternative hypotheses (16)

$$\begin{aligned} W &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} \frac{1 - 2\mu_0}{1 - \mu_0} + \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} \frac{\mu_0}{1 - \mu_0} + o_p(1) \\ &= \frac{(n - q_n)(T - \mu_0)}{\sqrt{\sigma_0^2}} + o_p(1). \end{aligned} \quad (\text{A.16})$$

It follows Proposition 1 that $\frac{(n-q_n)(T-\mu_0)}{\sqrt{\sigma_0^2}} \xrightarrow{d} N\left(\frac{\phi\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\tilde{\mathbf{\Gamma}} - \frac{\text{tr}(\tilde{\mathbf{\Gamma}})}{n-q_n} \mathbf{I}_{n-q_n}\right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{\sigma_0^2}}, 1\right)$. This, together with (A.16) yields that

$$W - \frac{\phi\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\tilde{\mathbf{\Gamma}} - \frac{\text{tr}(\tilde{\mathbf{\Gamma}})}{n-q_n} \mathbf{I}_{n-q_n}\right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{\sigma_0^2}} \xrightarrow{d} N(0, 1).$$

This completes the proof of (ii). $\square$

Proof of Corollary 1

$$\beta_{\text{BF}} = \Pr(T_{\text{BF}} \geq z_{1-\alpha})$$

$$\begin{aligned} &= \Pr\left(\frac{2 \log \text{BF}_{10} - \sum_{i=1}^{n-q_n} \log \gamma_i + \sum_{i=1}^{n-q_n} \log(\gamma_i + \lambda_i) + (n - q_n)}{\left(\log(1 - \mu_0) - \frac{\text{Var}_0(T)}{2(1-\mu_0)^2}\right) \sqrt{\sigma_0^2/(1 - \mu_0)}} \geq z_{1-\alpha}\right) \\ &= \Pr\left(\frac{2 \log \text{BF}_{10} - \sum_{i=1}^{n-q_n} \log \gamma_i + \sum_{i=1}^{n-q_n} \log(\gamma_i + \lambda_i) + (n - q_n)}{\left(\log(1 - \mu_0) - \frac{\text{Var}_0(T)}{2(1-\mu_0)^2}\right) \sqrt{\sigma_0^2/(1 - \mu_0)}} \right. \\ &\quad \left. - \frac{\phi\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\tilde{\mathbf{\Gamma}} - \frac{\text{tr}(\tilde{\mathbf{\Gamma}})}{n-q_n} \mathbf{I}_{n-q_n}\right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{\sigma_0^2}} \geq z_{1-\alpha} - \frac{\phi\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\tilde{\mathbf{\Gamma}} - \frac{\text{tr}(\tilde{\mathbf{\Gamma}})}{n-q_n} \mathbf{I}_{n-q_n}\right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{\sigma_0^2}}\right) \\ &= \Phi\left(-z_{1-\alpha} + \frac{\phi\beta_2^\top \tilde{\mathbf{V}}_2 \tilde{\mathbf{A}}_2^2 \left(\tilde{\mathbf{\Gamma}} - \frac{\text{tr}(\tilde{\mathbf{\Gamma}})}{n-q_n} \mathbf{I}_{n-q_n}\right) \tilde{\mathbf{V}}_2^\top \beta_2}{\sqrt{\sigma_0^2}}\right). \end{aligned} \quad (\text{A.17})$$

This completes the proofs. $\square$

Proof of Proposition 2 Let $m = n - q_n$, $\mu_{02} = \frac{2\text{tr}(\tilde{\mathbf{\Gamma}}^2) + (\text{tr}(\tilde{\mathbf{\Gamma}}))^2}{(n-q_n)(n-q_n+2)}$ and $\mathbf{A} = \mathbf{U}_1 \tilde{\mathbf{U}}_2 \in \mathbb{R}^{n \times m}$, which satisfies $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_m$. Writing $w = \mathbf{A}^\top \epsilon$, the statistic reduces to

$$\begin{aligned} T &= \frac{w^\top \tilde{\mathbf{\Gamma}} w}{w^\top w} = \frac{\sum_{i=1}^m a_i w_i^2}{\sum_{i=1}^m w_i^2}, \\ a_i &= \frac{\lambda_i}{\gamma_i + \lambda_i} \in [c, c'], \quad 0 < c < c' < 1. \end{aligned}$$

Since T is a weighted average of the a_i , we have $c \leq T \leq c'$ almost surely, thus $E(T^4) = O(1)$.

Let $\mu_0 = \bar{a}$ and $b_i = a_i - \mu_0$, $i = 1, 2, \dots, n$, so that $\sum_i b_i = 0$. Then

$$T - \mu_0 = \frac{Y}{D}, \quad Y = \boldsymbol{\epsilon}^\top M \boldsymbol{\epsilon}, \quad M = \mathbf{A} \operatorname{diag}(b_1, b_2, \dots, b_n) \mathbf{A}^\top, \quad D = \boldsymbol{\epsilon}^\top H \boldsymbol{\epsilon}, \quad H = \mathbf{A} \mathbf{A}^\top.$$

Using $\mathbf{A}^\top \mathbf{A} = \mathbf{I}_m$, we obtain $\operatorname{tr}(M) = 0$, $\operatorname{tr}(M^2) = \sum_i b_i^2 = \frac{1}{2} \sigma_0^2 = \Theta(m)$, and $\operatorname{tr}(H) = \operatorname{tr}(H^2) = m$. Define $\kappa' = \sum_i M_{ii}^2 = O(m)$. By the standard quadratic-form moment identities for i.i.d. errors, $E(Y) = 0$, $E(Y^2) = \sigma_0^2 + (\mu_4 - 3)\kappa' = \Theta(m)$, where $\mu_4 = E(\epsilon_1^4)$. The symmetry of the error distribution ensures that all odd cross-moments vanish, and the finite eighth moment guarantees $E(Y^4) = O(m^2)$ and $E\{(D - m)^4\} = O(m^2)$, which are needed to control the remainder terms below.

The Taylor expansion of $(T - \mu_0)^2 = Y^2/D^2$ around $D = m$, justified by the boundedness of T together with the moment bounds above, gives

$$E\{(T - \mu_0)^2\} - \sigma_0^2/(n - q_n)(n - q_n + 2) = \frac{(\mu_4 - 3)\kappa'}{m^2} (1 + o(1)) = \Theta(m^{-1}),$$

since the leading σ_0^2 terms cancel. Consequently,

$$E(R) = \frac{m[E\{(T - \mu_0)^2\} - \sigma_0^2/m(m + 2)]}{2\mu_0\sqrt{\sigma_0^2}} = \frac{(\mu_4 - 3)\kappa'}{2\mu_0 m\sqrt{\sigma_0^2}} (1 + o(1)) = O(m^{-1/2}).$$

Similarly, $\operatorname{Var}\{(T - \mu_0)^2\} = 2(E(Y^2))^2/m^4 (1 + o(1)) = \Theta(m^{-2})$, so that

$$\operatorname{Var}(R) = \frac{m^2}{4\mu_0^2\sigma_0^2} \operatorname{Var}\{(T - \mu_0)^2\} = O(m^{-1}).$$

Combining the two results, gives

$$E(R) = O(m^{-1/2}), \quad \operatorname{Var}(R) = O(m^{-1}),$$

and therefore

$$R = O_p(m^{-1/2}) \rightarrow 0 \quad \text{as } n - q_n \rightarrow \infty.$$

This completes the proofs. $\square$

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